

Review

Financial fraud detection using graph neural networks: A systematic review

Soroor Motie^{*}, Bijan Raahemi

Knowledge Discovery and Data Mining Lab, Telfer School of Management, University of Ottawa, 55 Laurier Avenue East, Ottawa ON K1N6N5, Canada



ARTICLE INFO

Keywords:

Graph neural networks
Anomaly detection
Financial fraud detection
Graph representation learning
GNNs

ABSTRACT

Financial fraud is a persistent problem in the finance industry that may have severe consequences for individuals, businesses, and economies. Graph Neural Networks (GNNs) are a class of deep learning models designed to operate on graph data structures that consist of nodes and edges connecting them. GNNs have emerged as a powerful tool for detecting fraudulent activities in complex financial systems because they can analyze the network structure of financial transactions, capturing patterns and anomalies that traditional rule-based and machine learning methods might miss.

The objective of this systematic review is to provide a comprehensive overview of the current state-of-the-art technologies in using Graph Neural Networks (GNNs) for financial fraud detection, identify the gaps and limitations in the existing research, and suggest potential directions for future research.

We searched five academic databases, including Web of Science, Scopus, IEEE Xplore, ACM, and science direct using specific keywords and search strings related to graph neural networks, financial areas, and anomaly detection to identify relevant publications, resulting in a total of 388 unique articles. We selected the relevant publications based on the inclusion, exclusion, and quality assessment criteria, and 33 articles were included in the review. In addition, forward snowballing was used to identify relevant papers that were not captured in the initial search. Data was extracted from the selected articles, then analyzed and summarized to identify current state, gaps, and trends in the literature.

Our review presents a new taxonomy of GNNs applied in financial fraud detection and identifies potential research directions in this field. We find that GNNs applied to financial fraud detection have mostly been employed in a supervised or semi-supervised manner, with limited exploration of unsupervised approaches. In addition to financial areas, we explore the different types of graphs such as homogeneous, heterogeneous, static, temporal, and dynamic graphs, and investigate the various learning mechanisms and anomaly types studied. We also note a lack of research on edge-level and graph-level anomaly detection commonly employed in financial domain.

1. Introduction

In recent years, the financial industry has witnessed a significant increase in the volume and complexity of financial transactions, leading to a surge in fraudulent activities. Financial fraud detection has become a critical concern for businesses, regulators, and consumers alike (Abdallah, Maarof, and Zainal, 2016). This recent rise in fraudulent activities underscores the crucial need for effective fraud detection. According to the CAFC (Canadian Anti-Fraud Centre) report, in 2022, more than 91,190 reports of fraud were recorded with 57,055 victims experiencing losses exceeding \$531 million. This highlights the fact that financial fraud can result in the loss of large amounts of money and undermine trust in financial institutions and systems. Financial fraud

detection may help prevent the unauthorized use of illegal or deceptive practices to obtain financial gain.

Fraud Detection Systems aim to detect fraudulent activities in the system and report them to system administrators through different techniques. Traditional rule-based systems and classical machine learning techniques have been employed to detect fraud (Hilal, Gadsden, and Yawney 2022); however, they have limitations in dealing with the growing intricacy of fraud patterns and massive amounts of data. Deep learning models, on the other hand, have demonstrated exceptional performance in various fields, such as computer vision, natural language processing, and speech recognition, due to their ability to learn intricate features and representations from raw data. This makes deep learning an attractive approach for financial fraud detection.

^{*} Corresponding author.

E-mail address: smoti088@uottawa.ca (S. Motie).

<https://doi.org/10.1016/j.eswa.2023.122156>

Received 14 June 2023; Received in revised form 25 September 2023; Accepted 12 October 2023

Available online 30 October 2023

0957-4174/© 2023 Elsevier Ltd. All rights reserved.

Traditional methods for fraud detection primarily rely on hand-crafted rules or feature engineering, which require domain expertise and are time-consuming to develop and maintain. Moreover, these methods often suffer from a lack of adaptability, making them ill-suited to address the evolving nature of financial fraud. Classical machine learning techniques, such as logistic regression, support vector machines, and decision trees, have also been employed to detect fraud (Hilal et al., 2022). Although these methods have shown some success, they struggle to handle high-dimensional data, non-linear relationships, and complex patterns that are common in financial fraud detection.

Deep learning models, particularly neural networks, offer a promising alternative for financial fraud detection due to their capacity to learn complex patterns and representations automatically from raw data. This eliminates the need for manual feature engineering, which is both time-consuming and domain specific. Furthermore, deep learning models can handle high dimensional data and model nonlinear relationships, making them well-suited for identifying subtle, hidden patterns indicative of fraudulent activities. Several deep learning architectures have been applied to financial fraud detection, including feedforward neural networks, recurrent neural networks (RNNs), convolutional neural networks (CNNs) (Hilal et al., 2022), and graph neural networks (GNNs). These models have been employed in various financial domains, such as credit card fraud, insurance, and money laundering. Deep learning models have been shown to outperform traditional methods in terms of accuracy, adaptability, and scalability.

Over the past ten years, graph-based fraud detection has gained prominence due to significant advancements in Graph Neural Networks (GNNs) and Graph Representation Learning (Niepert, Ahmed, and Kutzkov, 2016; Hamilton, Ying, and Leskovec, 2017). Graph neural networks apply representation learning involves mapping or embedding nodes or (sub)graphs into a low-dimensional vector space to detect anomalies at different levels, including nodes, edges, subgraphs, or entire graphs. In the financial domain, GNNs can be used to model relationships between customers, transactions, and merchants and trained to identify patterns of fraudulent activities such as money laundering and credit card fraud.

The graph used in financial fraud detection typically consists of nodes and edges representing accounts and transactions, respectively. Each node represents a financial account, such as a bank account, credit card account, or any other financial entity involved in transactions. Nodes may have associated attributes, such as account type, transaction history, account balance, account owner information, or other metadata relevant to the fraud detection task. Each edge represents a financial transaction between two accounts. The direction of the edge indicates the direction of the transaction (e.g., from one account to another). Edges may have associated weights, representing transaction amounts or other quantitative information (e.g., transaction timestamp).

By using message passing algorithms to propagate information between connected nodes, GNNs can learn representations that capture the structure of the graph and the relationships between its nodes. As the GNN processes the financial transaction graph and learns node embeddings, it simultaneously computes fraud scores for each account or transaction based on the learned representations and patterns. These fraud scores quantify the likelihood of an account or transaction being involved in fraudulent activities. The GNN applies a threshold to the computed fraud scores to distinguish between normal and suspicious instances. Accounts or transactions with fraud scores above a certain threshold are considered suspicious and flagged for further investigation. The threshold can be determined based on domain knowledge, the distribution of fraud scores, or by optimizing for a suitable trade-off between false positives and false negatives. The collaboration between the GNN's automated detection and human analysts' expertise ensures a robust and effective fraud detection system that helps financial institutions proactively combat financial fraud.

Through the lens of data type, transactional and textual data have been used typically for fraud detection (Ahmed, Mahmood, and Islam,

2016). However, the drawback of limiting data to text and transactional data is in not noticing relationships between entities in a network. In contrast, graphs (Akoglu,Tong,and Koutra, 2015), as a collection of nodes (e.g., customers) and edges (e.g., money transactions), can detect structural fraud by modeling the influence of each node not only on its neighbors, but also on the whole network.

This study provides a comprehensive review of the literature on using graph neural networks and graph representation learning for financial fraud detection. The scope of this review is structured into three distinct categories: data types, methods, and applications. From the data type perspective, the reviewed papers primarily involve graph data, focusing on graph representations over text or transactional data. However, some papers may encompass multi-modal data that includes graph representations. In terms of methods, the focus is limited to graph neural networks and graph representation learning approaches, while other machine learning and deep learning methods are not included. Application-wise, this review concentrates exclusively on financial fraud in the areas of taxation, banking, insurance, online payment transactions, and cryptocurrency. Other forms of fraud, such as computer network or telecommunication fraud, are beyond the scope of this review.

To the best of our knowledge, there is currently no systematic literature review that specifically explores the use of graph neural networks for financial fraud detection. While recent studies have conducted separate reviews on financial fraud detection techniques and graph anomaly detection, none have combined these areas of study to review the use of graph anomaly detection with graph neural networks for financial fraud detection.

For instance, (Ma et al., 2021) conducted a comprehensive survey on graph anomaly detection with deep learning methods but did not focus on graph neural networks (GNNs) specifically. (Kim et al., 2022) analyzed the current status and challenges of graph anomaly detection using GNNs but did not study its application in the financial sector. On the other hand, reviews centered on financial fraud detection, such as those conducted by (Cherif et al., 2023; Hilal et al., 2022; Zhu et al., 2021) broadly examine various techniques, but do not specifically mention the use of GNNs. Table 1 shows a comparison between existing surveys on anomaly detection.

Practitioners and researchers can use this systematic review as a comprehensive resource to understand the current state of the art in the application of Graph Neural Networks (GNNs) for financial fraud detection. Moreover, the review identifies gaps in the literature, which can guide future research in this area. This review can serve for practitioners and researchers who seek to enhance their decision-making processes when selecting appropriate methods for financial fraud detection using graphs, and to design new GNN-based methods that

Table 1
A comparison between existing surveys on anomaly detection.

Surveys	Graph Fraud Detection	Deep Learning	Financial Fraud Detection	Graph Neural Networks
Our Survey (Hilal et al., 2022)	✓	✓ ✓	✓ ✓	✓
(Kim et al., 2022)	✓	✓		✓
(Yang et al., 2022)		✓	✓	
(Yildiz et al., 2022)		✓	✓	
(Ma et al., 2021)	✓	✓	✓	
(Zhu et al., 2022)	✓	✓	✓	
(Akoglu et al., 2015b)	✓		✓	

address the identified research gaps.

The main contributions of this systematic review are as follows:

- Offering new taxonomy: A new taxonomy of graph neural networks for static graphs is proposed. Graph neural networks are categorized into four groups: Convolutional-based graph neural networks, Attention-based graph neural networks, Metapath-based graph neural networks, and Transformer-based graph neural networks.
- Identifying relevant resources: Abundant resources on graph neural networks, including state-of-the-art models, benchmark data sets, open-source codes, and financial applications, have been collected. This survey can be used as a reference for understanding, using, and developing different GNNs for various real-life financial applications.
- Identifying future directions: We identify areas where additional research is needed in the field of financial fraud detection using Graph Neural Networks. By identifying gaps and limitations in the current research, we can suggest potential future directions for research to improve the performance and applicability of GNNs in detecting financial fraud.

The rest of the paper is structured into five sections as follows: [Section 2](#) discusses the background information related to learning method of Graph Neural Networks and a number of preliminary definitions. [Section 3](#) provides an in-depth description of the systematic review, including the definition of research questions, selection of primary studies, source databases, search queries, exclusion and inclusion criteria, and quality assessment criteria. This section also includes the results of the search and PRISMA analysis. [Section 4](#) presents the data synthesis results of the review, including the answers to the two research questions. Finally, [Section 5](#) discusses the limitations and threats to the validity of the study. Further information is provided in the appendix.

2. Background

Here we will begin by introducing fundamental definitions that are crucial for comprehending the subsequent sections of the paper. Following that, we will delve into more information about Graph Neural Networks, aiming to provide a clearer understanding of this topic.

Definition 1. (Graph): Graph is a collection of nodes and edges that can be represented as $G = (V, E, X, M)$, where G denotes the graph, V is the set of nodes (vertices) with $|V| = n$, and E is the set of edges (connections) between the nodes with $|E| = m$. X is an $(n \times d)$ matrix representing the node features (with d being the dimension of the node feature vector), and M is an $(m \times p)$ matrix representing the edge features (with p being the dimension of the edge feature vector). Each row in the matrix X corresponds to a node's feature vector, while each row in the matrix M corresponds to a feature vector for an edge connecting a pair of nodes. It should be noted that in this review, the terms "static" and "homogeneous" graphs have same definition with the term "graph".

Definition 2. (Heterogeneous Graph): Heterogeneous graph consists of multiple node and edge types. In a heterogeneous graph, there are node type mapping function $\theta: V \rightarrow T_V$ and edge type mapping function $\xi: E \rightarrow T_E$, where T_V and T_E represent the sets of node and edge types, respectively. A heterogeneous graph can be denoted as $G_{he} = (V_{he}, E_{he}, X_{he}, M_{he}, \theta, \xi)$, where G_{he} is the heterogeneous static graph, V_{he} and E_{he} have the same meaning as in the graph, and X_{he} and M_{he} represent the node and edge features, respectively, in the heterogeneous context.

Definition 3. (Temporal Graph): A temporal graph is a graph where the node features or edge features change over time. A temporal graph can be represented as $G(t) = (V, E, X(t), M(t))$, where $G(t)$ denotes the graph at time t , V and E have the same meaning as in the graph definition, $X(t)$ is an $(n \times d)$ matrix representing the node features at time t , and $M(t)$ is an $(m \times p)$ matrix representing the edge features at time t .

Definition 4. (Dynamic Graph): A dynamic graph is a graph where the existence of nodes and edges is associated with specific time intervals. Mathematically, a dynamic graph can be denoted as $G_d = (V_d, E_d)$, where G_d is the dynamic graph, $V_d = \{(V, t_s, t_e)\}$ represents the set of nodes, and $E_d = \{(u, v, t_s, t_e)\}$ represents the set of edges. In this representation, v is a vertex of the graph, and t_s and t_e are the start and end timestamps for the existence of the vertex, respectively, with $t_s \leq t_e$. Similarly, u and v are vertices in V_d , and t_s and t_e are the start and end timestamps for the existence of the edge, respectively, with $t_s \leq t_e$. Graph Neural Networks (GNNs) are a variant of neural networks designed for machine learning tasks on graph datasets. GNNs aim to learn an embedding h_v for each node (v). Then, using this embedding, they can estimate the appropriate output, such as node class or a continuous value for regression tasks.

The GNNs propagation formula can be more abstractly expressed using the message-passing framework. This framework involves two primary steps: message passing and aggregation. The propagation formula can be broken down into the following stages:

1. Message Passing: Messages are computed for each node-edge pair $(v, u) \in E$, where E is the set of edges in the graph. The message function M takes the hidden states of the source node (h_u^t), target node (h_v^t), and possibly edge features (e_{uv}) as inputs and generates a message:

$$m_{uv}^{t+1} = M(h_u^t, h_v^t, e_{uv}) \quad (2.1)$$

2 Aggregation: Messages from all neighboring nodes are aggregated at each target node v . The aggregation function A combines the messages from the neighbors $N(v)$ into a single aggregated message:

$$a_v^{t+1} = A(\{m_{uv}^{t+1} : \forall u \in N(v)\}) \quad (2.2)$$

3. Update: The aggregated message a_v^t is used to update the hidden state of the target node v . The update function U takes the current hidden state of the node (h_v^t) and the aggregated message (a_v^t) as inputs and generates an updated hidden state:

$$h_v^{t+1} = U(h_v^t, a_v^{t+1}) \quad (2.3)$$

These steps are repeated for a predefined number of iterations (t), as depicted in [Fig. 1](#), enabling the GNN to capture information from more distant nodes in the graph. Various GNN architectures can be constructed by defining different message functions M , aggregation functions A , and update functions U ([Xu et al., 2019](#)). Many popular GNN architectures, such as Graph Convolutional Networks (GCN) and GraphSAGE, can be expressed in this message-passing framework by specifying appropriate M , A , and U functions.

To further elaborate, GNNs have been shown to be effective in solving various tasks in both supervised and unsupervised settings, such as node classification, edge classification, and graph classification. In the context of fraud detection, most of the papers in this survey focused on node classification tasks, however, there are also some studies that leveraged GNNs for edge classification ([Ye et al., 2022](#)).

1. Node Classification: Given a graph $G(V, E)$ with node features X_v and target labels Y_v for each node $v \in V$, the goal of node classification is to predict the label Y_v of each node based on the generated embeddings. After obtaining node embeddings h_v from the GNN, a classifier function C is applied to predict the target labels ([Cai, Zheng, and Chang, 2018](#)).

$$y_v^{pred} = C(h_v) \quad (2.4)$$

This classifier can be a simple linear transformation followed by a SoftMax activation for multi-class classification, or a sigmoid activation for binary classification.

2. Edge Classification: Given a graph $G(V, E)$ with node features X_v and target labels Y_{uv} for each edge $(v, u) \in E$, the goal of edge classification is to predict the label Y_{uv} of each edge based on the generated embeddings of connected nodes. After obtaining node embeddings h_u and h_v from the GNN, an edge representation function R is applied to compute edge embeddings ([Cai et al., 2018](#)).

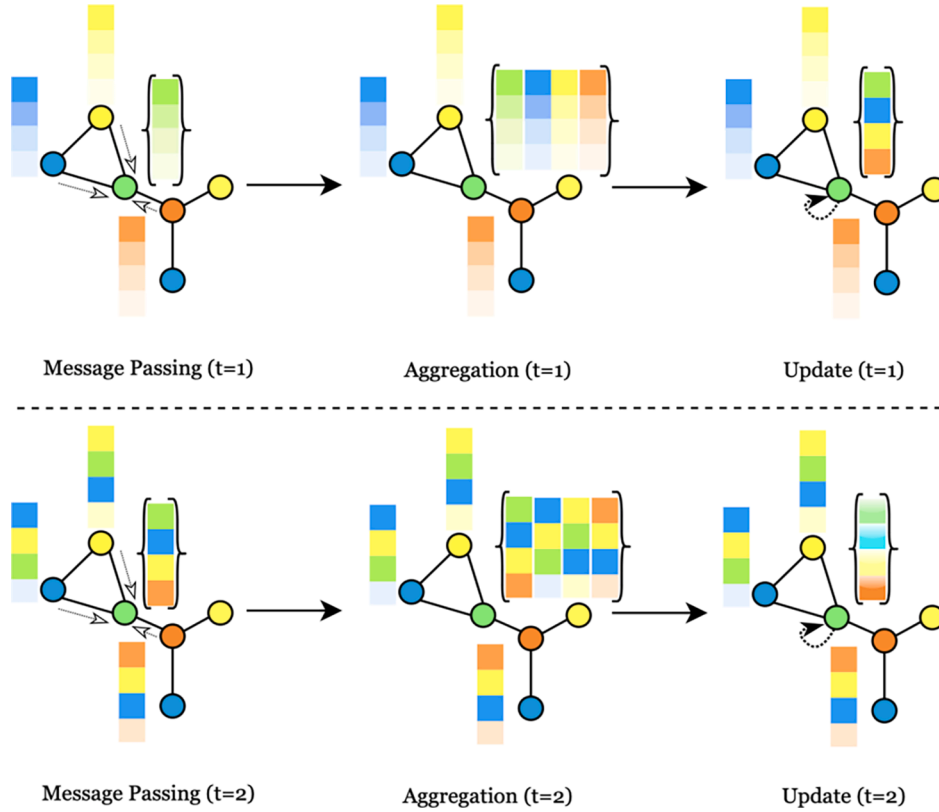


Fig. 1. Message Passing, Aggregation, and Update Process in Graph Neural Networks.

$$r_{uv} = R(h_u, h_v) \quad (2.5)$$

Various methods can be used for R , such as concatenation, element-wise addition, or element-wise multiplication. After obtaining edge embeddings r_{uv} , a classifier function C is applied to predict the target labels:

$$y_{uv}^{pred} = C(r_{uv}) \quad (2.6)$$

Similar to node classification, the classifier can be a linear transformation followed by a SoftMax or sigmoid activation.

3. Graph Classification: Given a dataset of graphs $\{G_i(V_i, E_i)\}$ with node features X_v and target labels Y_i for each graph G_i , the goal of graph classification is to predict the label Y_i of each graph based on the generated embeddings. After obtaining node embeddings h_v for all nodes in each graph, a graph-level aggregation function AG is applied to compute a graph embedding:

$$g_i = AG(\{h_v : \forall v \in V_i\}) \quad (2.7).$$

Various methods can be used for AG , such as sum, mean, max, or attention-based aggregation. After obtaining graph embeddings g_i , a classifier function C is applied to predict the target labels (Z. Zhang, Cui, and Zhu, 2022):

$$y_i^{pred} = C(g_i) \quad (2.8)$$

Similar to node and edge classification, the classifier can be a linear transformation followed by a SoftMax or sigmoid activation.

These formulations provide a general guideline for defining node, edge, and graph classification tasks using GNNs. Different GNN architectures and specific choices for aggregation, representation, and classifier functions can lead to a wide range of model variations for each task.

3. Methods

Following Kitchenham's approach (Kitchenham, 2004), we first

defined the research questions to identify the current status of research in financial fraud detection using graph neural networks. We then searched and selected relevant primary studies using pre-determined inclusion and exclusion criteria. Data was extracted from each study, and a meta-analysis was conducted to synthesize the results and identify trends and gaps in the literature. Accordingly, our approach consisted of the following four key steps:

- Definition of the research questions
- Selection of primary studies
- Data extraction
- Meta-analysis

3.1. Definition of research Question(s)

This study attempts to answer the following questions:

RQ1: What types of graph neural networks (GNNs) are proposed for anomaly detection in financial transactions?

RQ2: What are the current trends, gaps, and future research directions in the field of financial fraud detection using graph neural networks?

The first question helps us to compile and analyze the state-of-the-art graph neural network techniques in different financial areas. The second one guides us to identify trends of the research, gaps in the present studies, and the future direction of the research in this domain.

3.2. Selection of primary studies

The initial step in our research involved conducting an automated database search using a predefined query on five leading databases, including Web of Science, Scopus, IEEE Xplore, ACM, and science direct to identify relevant studies on the use of graph neural networks and graph representation learning for financial fraud detection. When duplicate records removed, then title and abstract screening, followed by full-text reviews, conducted to further eliminate irrelevant articles.

Additionally, the search strategy includes forward snowballing

Table 2

Search queries, keywords, and synonyms.

Main Concepts	Synonyms	Search Queries
Graph	graph	graph
Graph neural networks	neural networks neural network learning	"neural network*" OR learning
Fraud	fraud outlier anomaly	fraud OR outlier OR anomaly
Finance	finance financial tax credit money transaction transactional crypto Crypto-currency Insurance bank money	financ* OR tax OR credit OR money OR transaction* OR crypto* OR insurance OR bank

techniques by looking at the papers citing in included papers. Snowballing involves one-level forward chaining, in which citations from newer papers are traced.

3.2.1. Source databases

Our search string was conducted in five well-established databases, and grey literature sources will not be included in our search.

- Scopus¹
- Web of Science²
- IEEE Xplore³
- ACM Digital Library⁴
- Science Direct⁵

The databases Scopus, Web of Science, IEEE Xplore, and ACM Digital Library are among the most widely used and reputable academic databases in the field of computer science and engineering. Additionally, these databases offer advanced search features that enable precise and targeted searching to identify relevant articles for the research questions. While it is true that Science Direct's search options may not be as robust as other databases, based on prior experience there were a few relevant papers that are published exclusively on Science Direct. Therefore, it was important to include Science Direct in the search to ensure a comprehensive review of the literature.

3.2.2. Abstract search query

This review aims to provide a comprehensive review of the application of graph neural networks for anomaly detection for the purpose of financial fraud detection. Therefore, the search query focuses on three key concepts: graph neural networks, fraud detection, and the financial field. As shown in Table 2, the search query consists of these core concepts, with the addition of the term "graph" to ensure the data type is represented in the form of graphs.

To minimize the risk of missing relevant articles, synonyms and related terms for each concept were considered in developing the final search string. A combination of keywords and Boolean search operators

(i.e., "AND" and "OR") were used to identify a wide range of publications related to the use of graph neural networks in fraud detection in the financial sector.

The word 'graph' was split from 'neural network' in my search query for two reasons. First, by including the word 'graph' as a mandatory search term, I ensured the inclusion of papers related to graph neural networks. However, if I included the exact phrase 'graph neural network' in my search query, I would have missed out on important techniques such as graph convolutional neural networks or graph attention neural networks, which are large categories of graph neural network models. Therefore, I decided to split the term 'graph' from 'neural network' but also include the word 'graph' separately in my search query to ensure comprehensive coverage of graph-based techniques.

The term "network", as a synonym for graph, was deliberately omitted, since it would increase the number of false negative papers. Furthermore, the inclusion of the terms "representation learning" or "contrastive learning" in addition to the term "learning" was avoided as it did not significantly increase the number of relevant papers while decreasing the accuracy of the search results. Also, our objective was to use as many financial terms as possible to encompass the entire finance field within the scope of the review.

The final search queries in ACM, ScienceDirect, and IEEE Xplore were modified based on the limitations and capabilities of each database to ensure that the search query was optimized for each database. In the ACM database, the search was limited to the abstract of the papers, as the database did not support searching multiple parts of metadata simultaneously. In the ScienceDirect database, due to the limitations on the number of Boolean connectors allowed (max 8 per field), the query had to be modified and less critical elements omitted to ensure the inclusion of as many relevant papers as possible. The final search query for the IEEE Xplore database was constructed by specifying different metadata fields, including title, abstract, and keywords, before adding the relevant terms due to the complexity of searching through this database. The final search query in each database is presented in Table 3.

The search query will be performed on the following metadata fields.

Table 3

Final search queries on different databases.

Database	Search Query
Scopus, Web of Science, & ACM	("neural network*" OR learning) AND (fraud OR outlier OR anomaly) AND (financ* OR tax OR credit OR money OR transaction* OR crypto* OR insurance OR bank) AND (graph*)
ScienceDirect	(neural network OR learning) AND (fraud OR anomaly) AND (credit OR tax OR insurance) AND (graph OR network)
IEEE Xplore	((("Document Title":graph OR "Abstract":graph OR "Author Keywords":graph)) AND ((("Document Title":neural network OR "Abstract":neural network OR "Author Keywords":neural network OR "Document Title":learning OR "Abstract":learning OR "Author Keywords":learning)) AND ((("Document Title":fraud OR "Abstract":fraud OR "Author Keywords":fraud OR "Document Title":anomaly OR "Abstract":anomaly OR "Author Keywords":anomaly)) AND ((("Document Title":financ* OR "Abstract":financ* OR "Author Keywords":financ OR "Document Title":tax OR "Abstract":tax OR "Author Keywords":tax OR "Document Title":credit OR "Abstract":credit OR "Author Keywords":credit OR "Document Title":money OR "Abstract":money OR "Author Keywords":money OR "Document Title":transaction* OR "Abstract":transaction* OR "Author Keywords":transaction* OR "Document Title":crypto* OR "Abstract":crypto* OR "Author Keywords":crypto* OR "Document Title":insurance OR "Abstract":insurance OR "Author Keywords":insurance OR "Document Title":bank OR "Abstract":bank OR "Author Keywords":bank)))

¹ <https://www.scopus.com/>.

² <https://www.webofknowledge.com/>.

³ <https://www.ieeexplore.ieee.org/>.

⁴ <https://dl.acm.org/>.

⁵ <https://www.sciencedirect.com/>.

- Title
- Abstract
- Authors' Keywords

3.2.3. Exclusion criteria

This review has excluded articles with the following criteria:

- Papers that do not have the financial area as their main focus of research (i.e., some papers may mention financial fraud detection as one of several applications in a broader field but not specifically focus on this area).
- Papers that do not have GNNs as their main method of research (i.e., some papers may mention learning, but they refer to machine learning and deep learning methods other than GNNs).
- Papers that employ graph neural networks in finance but for broader tasks, not specifically for anomaly or fraud detection.
- Papers that are not peer-reviewed, including preprinted papers, posters, short papers, abstracts, reports, thesis, and book chapters.
- Articles not written in English.
- Duplicate publications with the same authors, with the latest one selected.

3.2.4. Inclusion criteria

The current review includes academic peer-reviewed conferences and journal articles that employ graph neural networks as their method and apply them for financial anomaly detection. The time period covered by this review is not restricted, as the concept of graph neural networks is relatively new (Table 4).

3.2.5. Quality assessment criteria

The quality assessment criteria for each article are as follows.

- The contributions of the study must be effectively presented through conducting experiments.
- The methodology must be clearly and convincingly demonstrated in the article.

Articles that fail to meet the quality criteria will be excluded from the review. The Covidence software has been utilized to clearly employ the exclusion and quality assessment criteria and to enforce them during the review process.

3.2.6. Search results and PRISMA results

The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guideline (Page et al., 2021) is a widely recognized set of evidence-based recommendations for the reporting of systematic reviews and meta-analyses. It was first published in 2009 and has since been updated several times to reflect the latest best practices in research reporting.

The flow diagram in Fig. 2 provides a visual representation of the systematic review process, from the initial identification of relevant

studies to the final inclusion and exclusion of studies. The total number of 578 studies were extracted, resulting in the identification of a total of 388 unique articles from. Following title and abstract screening, 60 articles were deemed suitable for full-text screening. After a rigorous full-text screening process, 30 articles were found to meet the pre-defined inclusion and exclusion criteria for the review. In addition, the snowballing technique was applied, which resulted in the identification of three additional relevant articles. Overall, a total of 33 articles were included in the review, which provided a comprehensive and robust overview of the topic under investigation.

3.3. Data extraction

In the data extraction phase of this study, relevant data and information from the selected papers were collected to address the research questions. The information collected in Table 5 through full-text screening serves to identify the relationships between different columns and uncover any hidden insights to answer the research questions.

The scope of graph neural networks examined not only the models used but also the input data, learning mechanisms, performance, and anomaly detection methods. Moreover, information on the finance area and the GNN used was extracted. Data accessibility and code availability information were also noted. Other information obtained from the initial automated search, such as the publication year and the type of publication (journal or conference), was used to identify publication trends, which could be analyzed using the publication year. This collection of data was employed in answering our two research questions on current GNN models and potential gaps.

3.4. Meta analysis

Meta-analysis is a statistical method used in research to combine the results from multiple studies. Meta-analysis can be a useful tool in synthesizing the findings of multiple studies and drawing conclusions that may not be possible based on individual studies alone. This section demonstrates the primary investigation results of the selected articles using a statistical approach to show the distribution of selected papers over years and based on their publication type as shown in Fig. 3.

The number of papers published on the topic of graph neural networks for financial fraud detection has increased significantly over the years, indicating a growing interest in this area of research. This trend is particularly evident from 2019 to 2023, indicating that this is a relatively new and emerging topic. Furthermore, the increase in the number of journal papers over time suggests that the research is becoming more established and gaining recognition in the academic community. We will answer the defined research questions based on the ultimately determined papers (47 papers) in the data synthesis results section.

4. Data synthesis results

The data synthesis results section provides a detailed analysis of the answers to the two research questions. The first research question focuses on the types of graph neural networks (GNNs) proposed for anomaly detection in financial transactions. The section presents a thorough discussion of the different types of GNNs, including graph convolutional networks (GCNs), graph attention networks (GATs), and others. The section also examines the performance of these GNNs on different financial fraud datasets, along with the corresponding financial areas, source, availability, and other relevant details. The second research question investigates the current trends, gaps, and future research directions in the field of financial fraud detection using GNNs. The section provides a comprehensive analysis of the literature to identify emerging trends, gaps, and challenges in the field. It also suggests potential future research directions to address the identified gaps and challenges.

Table 4
Search Results on 5 different databases.

No	Database	Web Address	Number of Papers
1	Scopus	https://www.scopus.com/	306
2	Web of science	https://www.webofknowledge.com/	127
3	IEEE Xplore	https://www.ieeexplore.ieee.org/	71
4	ScienceDirect	https://www.sciencedirect.com/	40
5	ACM Digital Library	https://dl.acm.org/	34
Sum Including Duplications			578
Number of duplicates			190
Sum Removing Duplications			388



Fig. 2. PRISMA Guideline Results.

Table 5
Data extraction criteria.

Data Category	Data Column	Description	Extracted Data Instances
Publication Data	Publication Year	Publication Year	From 2019 to 2023
	Document Type	Where Article Published	Journal or Conference
Learning Mechanism	GNN Model	Applied GNN models	GCN, GraphSage, GAT, etc.
	Learning Method	Training Approach	Supervised, Semi-supervised, or Unsupervised
Input Data	Graph Type	Type of Input Graph	Static, Temporal, or Dynamic
	Heterogeneity	Different Types of Nodes or Edges	Homogeneous or Heterogenous
	Nodes	What Nodes Represent	e.g. Users
	Edges	What Nodes Represent	e.g. Users who Have Shared Delivery Address, or Phone or Device, Have Connection
	# Nodes	Number of Nodes	e.g. 25,257
	# Edges	Number of Edges	e.g. 50,422
	Features	Node or Edge Features	e.g. timestamps, Bitcoin fees, and volume
Anomaly Data	Anomaly Definition	What Counts as Anomaly	e.g. Illicit Bitcoin Addresses
	Anomaly Level	Level of Anomaly	Node Level or Edge Level
Performance	Performance	GNN Model Performance	Accuracy = %92.4
Application	Financial Area	Financial Area	Taxation, Insurance, Banking, etc.
Producibility	Codes	Github Link	github.com/WangXuhong CN/APAN
Datasets	Datasets	Source of Data	xblock.pro/Ethereum
	Datasets Access	Availability of Data	Public or Private

4.1. RQ1: Graph neural networks (GNNs) applied for financial fraud detection

GNNs have been applied in various ways to enhance fraud detection capabilities, from modeling intricate transaction networks to learning robust representations of suspicious activities. By enabling information propagation across interconnected nodes through message-passing and

graph convolutions, GNNs have proven adept at capturing the context and relationships critical for distinguishing legitimate financial behaviors from fraudulent ones.

Graph Neural Networks (GNNs) are utilized in financial fraud detection because of their ability to effectively capture the complex and interconnected relationships within financial networks. GNNs offer the advantage of modelling complex relationships over traditional methods and other existing techniques, making them particularly well-suited for fraud detection in the financial domain. Financial transactions and activities form intricate networks with numerous interdependencies. GNNs can naturally handle such complex structures by propagating information across interconnected nodes. This enables GNNs to capture not only direct connections but also higher-order relationships, providing a more comprehensive view of fraudulent patterns. Also, GNNs have the capacity to learn non-linear representations of data. This is crucial in financial fraud detection, where fraudulent behavior might not follow simple patterns. GNNs can identify subtle and non-obvious relationships, which might be missed by linear models or rule-based systems.

Graph Neural Networks can detect financial fraud by leveraging the graph representation of financial transactions and accounts. By walking through an example of how GNNs can be used to detect fraudulent activities related to money laundering, we can elaborate the concept. In the example of money laundering, each account becomes a node in the graph, and transactions between accounts create directed edges with transaction amounts as edge weights. The graph contains information about the transaction amounts, timestamps, and the sender and receiver accounts. GNNs will be applied to learn node embeddings of the accounts in the graph. GNNs propagates information through the graph, capturing patterns and relationships between accounts.

GNNs can detect unusual transaction patterns that might indicate money laundering. For instance, it may identify cycles of transactions between accounts, where funds are continually transferred in a circular manner to obfuscate the origin of the money. GNNs also can identify accounts with anomalous behavior in terms of transaction volume, frequency, or amounts. Accounts that deviate significantly from the typical behavior of legitimate accounts might raise suspicion of money laundering.

In clustering point of view, GNNs group similar accounts together based on their embeddings. Clusters of accounts with similar transaction patterns could indicate potential money laundering networks or entities. GNNs can analyze the flow of funds between accounts. They may identify accounts that frequently receive funds from a wide range of unrelated accounts and subsequently transfer those funds to other seemingly unrelated accounts, which could suggest money laundering

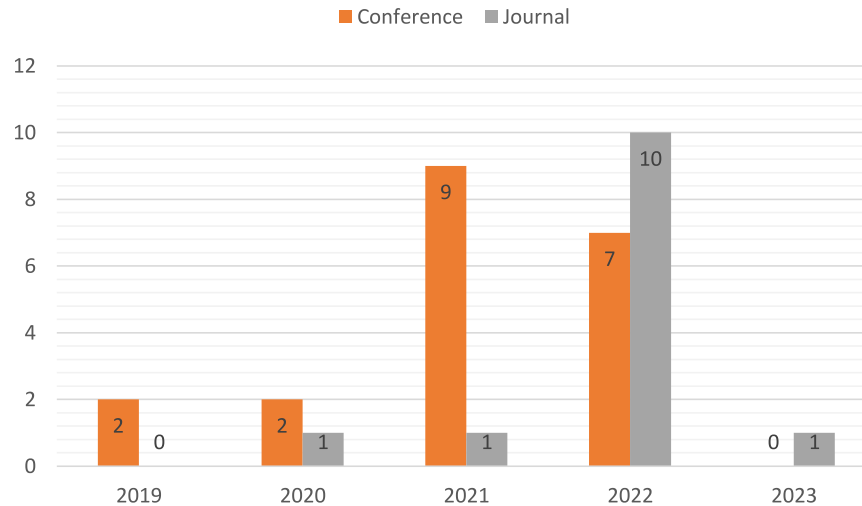


Fig. 3. Number of Document Types (Journal or Conference) per Year.

activities. Incorporating temporal information, GNNs can detect sudden changes in transaction patterns over time. A sudden increase in transaction volume or changes in transaction destinations might be indicative of money laundering attempts.

The performance of GNNs in financial fraud detection is often evaluated in terms of accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC). GNNs have demonstrated superior performance compared to traditional machine learning models or deep learning methods, in some cases, especially when the fraud patterns involve intricate relationships or interactions between accounts. However, the relative performance of GNNs depends on factors such as the size of the financial graph, the quality of the data, the selection of features, and the availability of labeled data. In certain cases, traditional

machine learning models (e.g., logistic regression, random forests) or deep learning methods (e.g., feedforward neural networks, recurrent neural networks) might still perform competitively, particularly in cases where the data is not graph-structured or when the graph is relatively small. In practice, combining different methods, including GNNs, traditional machine learning models, and deep learning techniques, in an ensemble framework can often lead to improved overall performance in detecting financial fraud.

There is no unique method proposed in the literature for categorizing various GNN models. For example, [Zhou et al. \(2020\)](#) categorizes GNNs based on their computational modules, including pooling, propagation, and sampling modules. [Wu et al. \(2021\)](#) categorizes GNNs as recurrent, convolutional, graph autoencoder, and spatial-temporal GNNs. [Y. Zhou](#)

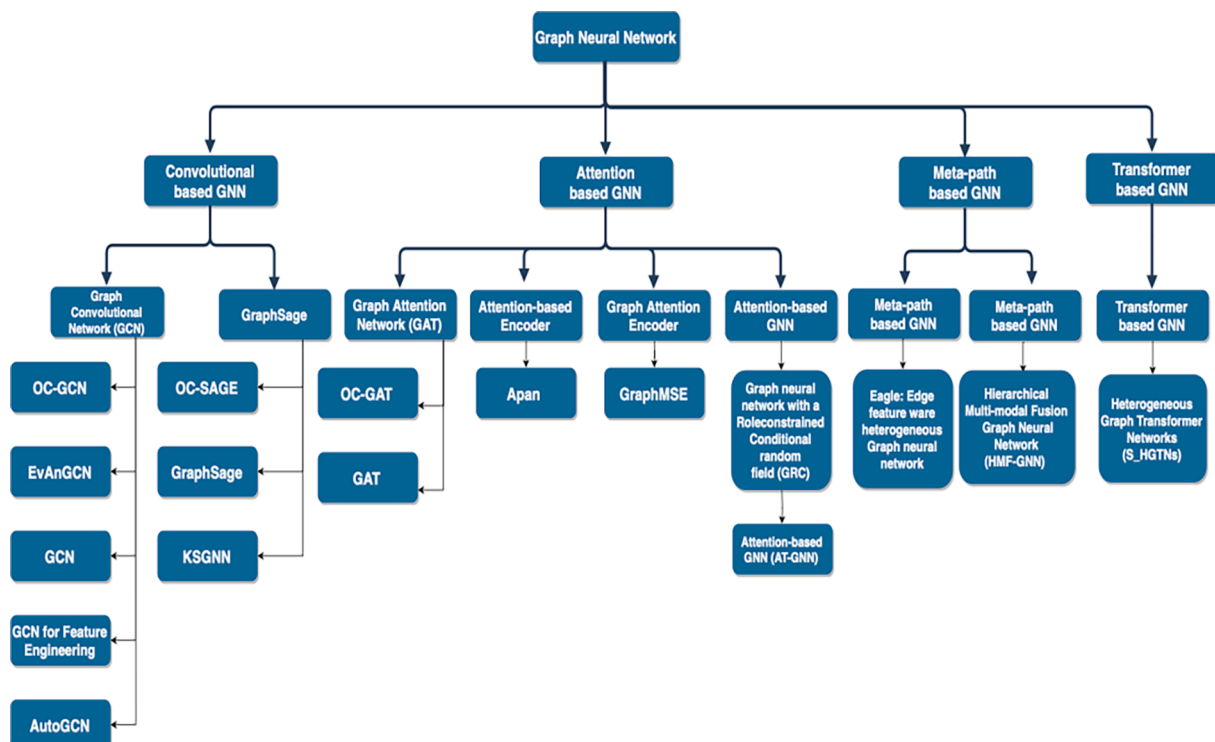


Fig. 4. GNN Models Categorization.

et al. (2022) categorizes GNNs based on their fundamental architecture, including graph convolutional neural networks, graph pooling operators, graph attention mechanisms, and graph recurrent neural networks.

Our review introduces a novel taxonomy of Graph Neural Networks used in financial fraud detection, Fig. 4. We have examined models in static graphs and graphs that change over time, including temporal and dynamic graphs. We have categorized models into Convolutional-based, Attention-based, Metapath-based, and Transformer-based models, based on our analysis of existing literature in the field. The four categories are as follows:

Convolutional-based models: These GNNs use a convolutional operation to update node representations based on the features of the node and its local neighborhood. These are the most basic type of GNNs.

Attention-based models: These GNNs utilize an attention mechanism alongside convolutional layers to allocate varying weights to different nodes or edges, enabling the model to learn to focus on specific nodes or edges that are more relevant to the task, while ignoring irrelevant information.

Metapath-based models: These GNNs use *meta*-paths, which are sequences of node types and edge types, to capture the semantic meaning of the graph structure. These models add domain-specific knowledge to GNNs and help to generate better node representations.

Transformer-based models: Compared to metapath-based models, these models equipped GNNs with learned metapaths rather than predefined paths. These group of models can identify useful connections between nodes on the original graph, while learning effective node representation.

4.1.1. Convolutional based GNNs

Convolutional based graph neural networks leverage the graph convolution mechanism, which is one of the most widely used information aggregation techniques in graph neural networks. The fundamental concept behind this mechanism is to employ convolutional operations on the graph structure to extract higher-level representations for each node (Fig. 5). These models are inspired by Convolutional Neural Networks (CNNs) but are independent of the node order (Xiao et al., 2022).

Based on the findings presented in Table 6, two types of Convolutional-based GNNs have been identified as being used for financial fraud detection on static graphs: Graph Convolutional Networks (GCNs) and GraphSage.

Graph Convolutional Networks (GCN)

In 2016, Kipf and Welling presented Graph Convolutional Networks (GCNs) which use a specialized type of convolutional neural network that utilizes graph convolutional operators to learn feature representations for individual nodes within a graph. The graph convolutional operators allow the network to aggregate information from a node's neighbors in order to learn more complex and informative representations of the overall graph structure (Kipf and Welling, 2017).

The propagation rule in this multi-layer Graph Convolutional Network (GCN) is based on:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right) \quad (2.9)$$

here, $\tilde{A} = A + I_N$ is the adjacency matrix of the undirected graph G with added self-connections. I_N is the identity matrix, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ and $W^{(l)}$ is a layer-specific trainable weight matrix. $\sigma(\cdot)$ denotes an activation function, such as the $ReLU(\cdot)$. $H^{(l)} \in R^{N \times D}$ is the matrix of activations in the l th layer; $H^{(0)} = X$.

GraphSage

GraphSAGE (Sample and Aggregate) is a graph convolutional network introduced by (Hamilton et al., 2017) that utilizes a different sampling and aggregation approach. Unlike graph convolutional networks (GCNs), GraphSAGE is designed to enable models to become inductive and handle previously unseen nodes. It achieves this by leveraging both the feature information of nodes and the structural attributes of a node's local neighborhood to learn a higher representation for each node.

4.1.2. Attention based GNNs

Attention-based GNNs are a class of GNNs that utilize the attention mechanism, which allows the model to focus on the most relevant nodes or edges in a graph during the message-passing process. In these models, the weights for the aggregation of information are learned dynamically based on the similarity between the nodes or edges, rather than being fixed as in graph convolutional networks.

Graph Attention Network (GAT)

Graph Attention Network (GAT) is a specific type of attention-based GNN proposed by Velickovic et al. (2018) as a type of neural network architecture that employs self-attention mechanism, which allows it to selectively aggregate information from different neighbors depending on their importance to the node and model complex relationships and dependencies within graph data more effectively (Velicković et al., 2018). The computation of attention coefficients (α) and hidden states (h) of each node is carried out through:

$$\alpha_{ij} = \frac{\exp \left(\text{LeakyReLU} \left(\vec{a}^T \left[W \vec{h}_i \| W \vec{h}_j \right] \right) \right)}{\sum_{k \in N_i} \exp \left(\text{LeakyReLU} \left(\vec{a}^T \left[W \vec{h}_i \| W \vec{h}_k \right] \right) \right)} \quad (2.10)$$

$$\vec{h}'_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} W \vec{h}_j \right) \quad (2.11)$$

where W is the weight matrix associated with the linear transformation, which is applied to each node, and a is the weight vector of a single-layer MLP (Fig. 6).

4.1.3. Metapath based GNNs

A *meta*-path-based graph neural network is a unique approach to handling heterogeneous graphs, focusing on the relationships between different types of nodes and edges (Sun et al., 2011). To define a *meta*-path, a schema is employed for a heterogeneous graph, which maps its nodes and edges to corresponding object types. Subsequently, a path is defined in this new space, resulting in a *meta*-path. For example, in a citation network, one could define a co-authorship relation as a *meta*-path. In this context, a *metapath* is a sequence of nodes and edges that should be traversed in order to connect two nodes of interest (Yi Li et al., 2021) (Fig. 7).

In a *meta*-path-based Graph Neural Network, node embeddings are updated based on the *meta*-path instances associated with a specific node. This process allows the model to capture complex relationships and patterns within the heterogeneous graph, improving its ability to analyze and understand the underlying data. By leveraging exogenous *meta*-paths, the graph neural network can more effectively process and learn from heterogeneous graphs, which often contain diverse and interconnected information.

4.1.4. Transformer based GNNs

The Graph Transformer Network (GTN) is a specialized Graph Neural Network designed for learning node representations in heterogeneous graphs (Yun et al., 2019). Unlike the previously described *meta*-path-based GNN, which focuses on updating node embeddings based on specific *meta*-path instances, GTN offers a more flexible approach to handling heterogeneous graphs. It achieves this by transforming a heterogeneous graph into multiple new graphs defined by *meta*-paths with

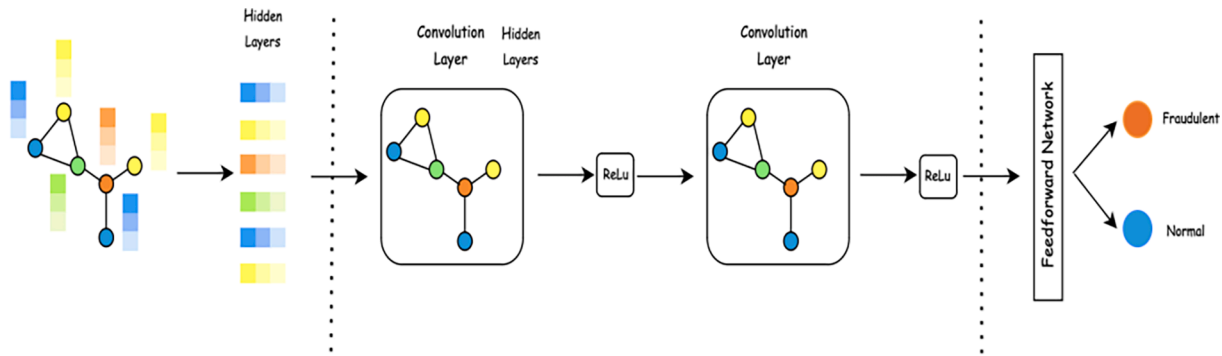


Fig. 5. Convolutional Based GNNs.

arbitrary edge types and lengths, using multiple Graph Transformer layers.

These layers work in tandem to learn new graph structures through the composition of adjacency matrices. This allows GTN to learn arbitrary *meta*-paths concerning edge types and path lengths, offering a more comprehensive understanding of the relationships within heterogeneous graphs. Additionally, GTN is designed to learn both short and long *meta*-paths, including original edges, making it suitable for applications where diverse *meta*-path lengths are important (Fig. 8).

Table 6 in our paper provides a categorization of Graph Neural Networks (GNNs), along with the corresponding models and references reviewed in our study. Some categories like convolutional and attention based GNNs are more explored than *meta*-path and transformer based GNNs, and some models like Graph convolutional networks, Graphsage, and Graph Attention Network are mostly employed even in their simple form.

4.2. RQ2: Current Trends, Gaps, and future research directions

In this section, we address the research question “What are the current trends, gaps, and future research directions in the field of financial fraud detection using graph neural networks?” by examining financial fraud detection using GNNs from various perspectives. Specifically, we analyze the application of GNNs in different financial domains such as cryptocurrency, online payment transactions, taxation, medical insurance, and credit cards. We also investigate the datasets used in previous studies, including their corresponding financial areas, source, and availability. Additionally, we explore the use of heterogeneous graphs, learning mechanisms, anomaly level, anomaly percentage, and the number of nodes and edges in GNN-based financial fraud detection. By considering these different factors, we aim to identify the current trends, gaps, and future research directions in this field.

4.2.1. Financial areas

Financial fraud detection is the process of identifying and preventing fraudulent activities in financial transactions. There are several methods used for fraud detection, including rule-based methods, machine learning, and deep learning techniques, including graph neural networks. In our review, based on existing papers, we have categorized financial fraud detection applying graph neural networks into five different categories: cryptocurrency, credit card, online transactions, taxation, and medical insurance. In Fig. 9, the proportion of papers published in each financial area reviewed in this study is illustrated.

For each category, researchers have constructed financial networks based on different definition of nodes and edges. Also the clear definition of fraud detection task and the goal of fraud detection in each category might differ. In this section, we elaborate on how researchers define different components of financial network in each category and how they define fraud detection in the specified task for each category.

For more detailed information on this topic, we refer to Table A2 in the appendix.

Cryptocurrency fraud detection

Cryptocurrency fraud detection refers to the use of various techniques and algorithms to identify and prevent fraudulent activities in cryptocurrency transactions. Cryptocurrencies such as Bitcoin and Ethereum have gained widespread adoption, making them a target for fraudulent activities such as money laundering, theft, and market manipulation. To combat these activities, researchers and practitioners have developed various approaches to detect fraud, including graph-based methods, machine learning, and deep learning techniques. The goal of cryptocurrency fraud detection is to protect investors, prevent market manipulation, and promote a safe and transparent financial environment for all cryptocurrency users.

Fraudulent transactions can compromise the security of blockchain networks, making it crucial to detect and prevent such anomalous behavior. Over 50 % of the reviewed papers in this study are devoted to fraud detection in cryptocurrencies, particularly Bitcoin and Ethereum. These papers share a common definition of nodes, edges, and fraud, and we will examine them to gain a deeper understanding of how these concepts are defined in the context of cryptocurrency. This emphasis on cryptocurrency may highlight its increasing significance in the financial industry and the need for effective fraud detection techniques in this field or this portion of papers may indicate the availability of data in cryptocurrency fraud detection compared to other financial fields.

In the majority of papers focused on fraud detection in Ethereum, the nodes of the graphs are the addresses of the senders and receivers, which represent accounts. The edges of the graph are unique pairs of sender and receiver accounts involved in a transaction, representing a relationship between the two nodes. Anomalies in these graphs are defined as phishing accounts or addresses on the Ethereum network.

Patel et al. conducted two studies on Ethereum fraud detection, using the same dataset. Their 2020 study focused on static graphs, while their 2022 study explored dynamic graphs. In both studies, the nodes were defined as addresses, and the edges of the graph were represented by unique pairs of sender and receiver accounts involved in a transaction. The researchers defined fraud as anomalous addresses or accounts that exhibited behavior deviating from the norm.

Similarly, Tan et al. (2021), Zhou et al. (2022) and Li et al. (2021) also defined nodes, edges, and frauds in the same way for Ethereum fraud detection. In contrast, Liu et al. (2022) investigated fraudulent smart contracts in Ethereum, where the nodes represent accounts, states, gas and fees, transactions, and blocks, and the edges represent relations between nodes in a heterogeneous graph.

Bitcoin transactions are typically represented as nodes in fraud detection studies, while edges represent payment flows from the source to the destination. Anomalies are defined as illicit transactions within the Bitcoin transaction graph, as outlined in several papers including (Alarab, Prakoornwit, and Nacer 2020; Zheng, 2022; Li et al., 2022; Han

Table 6
Graph Neural Networks Categorization.

GNN Category	GNN Model	Model Name	References
Convolutional based GNN	Graph Convolutional Network (GCN)	OC-GCN	(Patel et al., 2020)
		EvAnGCN	(Patel et al., 2022)
		GCN	(Tan, Tan, Zhang, & Li, 2021; Han et al., 2022; Li et al., 2022; Duan, Yan, Dong, Zhang, & Yu, 2022; Sergadeeva, Lavrova, & Zegzhda, 2022; Alarab et al., 2020; Reddy et al., 2021; Xia et al., 2022; Cheng et al., 2022; Chen et al., 2022; Singh et al., 2021; Wang et al., 2021)
	GraphSage	GCN for Feature Engineering	(Jing et al., 2021; Fu, Yu, & Feng, 2022)
		AutoGCN	(Lv et al., 2019)
		OC-SAGE	(Patel et al., 2020)
Attention based GNN	Graph Attention Network (GAT)	GraphSage	(Li et al., 2022; Han et al., 2022; Jing et al., 2021; Duan et al., 2022; Van Belle et al., 2020; Jing et al., 2019; Li et al., 2021; Van Belle et al., 2022)
		Knowledge-Guided Semi-supervised Graph Neural Network (KS-GNN)	(Rao et al., 2021)
		OC-GAT	(Patel et al., 2020)
	Attention-based Encoder	Graph Attention Network (GAT)	(Han et al., 2022; Liangguang et al., 2021; Zheng et al., 2022; Singh et al., 2021)
		Apan: Asynchronous propagation attention network	(Wang et al., 2021)
		GraphMSE	(Zhou et al., 2022)
	Attention-based GNN	Graph neural network with a Role-constrained Conditional random field (GRC)	(Xu et al., 2021)
		Attention-based GNN(AT-GNN)	(Tian et al., 2021)
		Eagle: Edge feature Aware heterogeneous Graph neural network	(Shi et al., 2023)
Meta-path based GNN	Meta-path based GNN	Hierarchical Multi-modal Fusion Graph Neural Network (HMF-GNN)	(Zhang et al., 2022)
		Path-based Graph Neural Network	
		Transformer-based GNN	
Transformer based GNN	Transformer-based GNN	Heterogeneous Graph Transformer Networks (S_HGTNs)	(Liu et al., 2022)

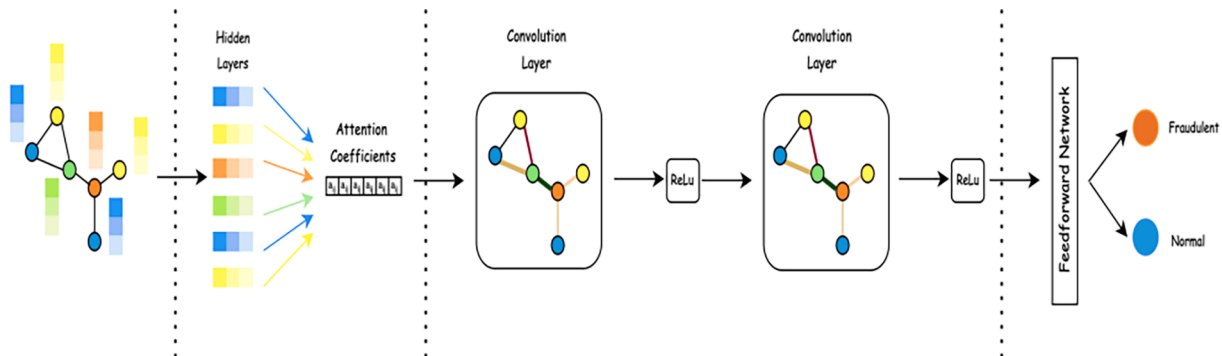


Fig. 6. Attention Based GNNs.

et al., 2022; Singh et al., 2021). In contrast, (Liangguang and Zhen, 2021) represents anonymous traders as nodes in the network, with each edge representing an anonymous, and fraud is defined as anomalous transactions or edges. In (Tian et al., 2021), nodes are defined as bitcoin addresses, and edges represent transactions. Anomalies in this study are defined as illicit Bitcoin addresses.

Credit Card Fraud Detection

Credit card fraud detection involves identifying fraudulent activities related to credit card transactions. It is a critical task in the financial industry, as credit card fraud can lead to significant financial losses for both the cardholder and the card issuer. Various techniques have been employed to detect credit card fraud, including rule-based systems, statistical models, and machine learning algorithms. In recent years, deep learning methods, such as neural networks and graph-based approaches, have also been explored for credit card fraud detection.

Jing et al. have conducted two studies in credit card fraud detection. In both studies, nodes were defined as credit card records or clients. However, in their 2019 study, edges represented the similarity degree of the two clients, while in their 2021 study, edges were extracted through

a GNN model in a network. In Cheng et al.'s work, they calculate the probability of fraudulent trades in a network where nodes represent users and merchants, and edges represent connections between a user and a merchant.

Van Belle et al. have also proposed two papers in credit card fraud detection. They differ in defining transactions in nodes, in addition to merchants and cardholders to include the embedding of transactions in a model. The edges represent connections between merchants and cardholders through the transaction node. Anomalies are defined as fraudulent transactions. Two recent studies by Rao et al. (2021) and Sergadeeva et al. (2022) focused on banking fraud detection using a typical approach where the nodes represent credit card users, and the edges represent their transactions. The goal is to detect fraudulent users by analyzing their transaction patterns and identifying anomalies.

Online Payment Transactions Fraud Detection

In recent years, the rise of e-commerce and online payment systems has led to an increase in fraudulent activities, such as identity theft, unauthorized transactions, and account takeovers. Online payment transactions fraud detection is the process of identifying and preventing

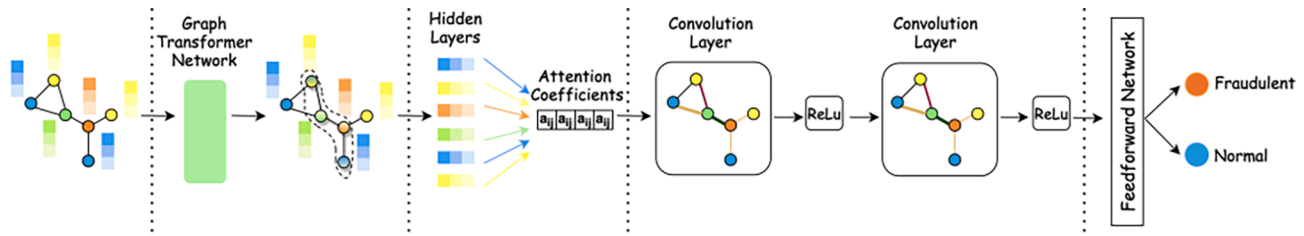


Fig. 7. Meta-path Based GNNs (Dotted lines indicates exogenous meth path).

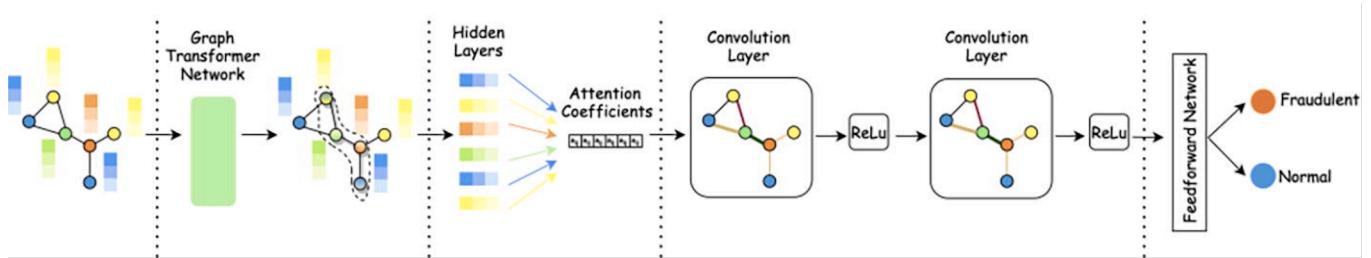


Fig. 8. Transformer Based GNNs (Dotted lines indicates meta path extracted from transformer).

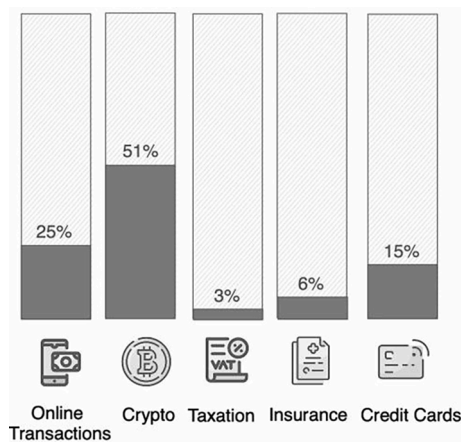


Fig. 9. Financial Areas that Applied Graph Neural Networks for Fraud Detection.

fraudulent activities that occur during online payment transactions. In most of online payment transactions, nodes are defined as users, except in two studies on heterogeneous graphs, including the work of Xu et al. (2021) that nodes are applicants, sellers, intermediaries and other users, and the study of Reddy et al. (2021) that nodes represents users and merchants.

Edges are represented differently in studies; money transferring (Wang et al., 2021; Reddy et al., 2021), relation between the applicant and other users (Xu et al., 2021), sharing delivery address, or phone or device (Lv et al., 2019), and lending money from institutions (Sergadeeva et al., 2022). Nodes may be fraudulent in all the studies except the of Wang et al. (2021) in which each edge has a label that indicates that whether an interaction is fraud. In addition, in Xia et al.'s study, they focused on money laundering applications in a graph where nodes represent transactions, and edges imply the directed payment flow between different transactions. Fraud is defined as illicit money laundering transactions.

Taxation Fraud Detection

Taxation fraud detection refers to the use of data analysis and machine learning techniques to identify individuals or companies that engage in fraudulent activities related to taxes. This can include activities such as underreporting income, claiming false deductions, and hiding assets to avoid paying taxes. The goal of taxation fraud detection

is to improve tax compliance and deter fraudulent behavior, ultimately leading to a fairer and more effective tax system.

The paper by Shi et al. (2023) proposes an edge feature aware heterogeneous graph neural network model for tax evasion detection. The nodes in the graph represent production, investor, event, and company, and the edges represent various relationships between these nodes, including CC, CP, PC, PP, CE, CI, and II. The objective is to detect companies that are involved in tax evasion.

Medical Insurance Fraud Detection

Medical insurance fraud detection is the process of identifying and preventing fraudulent activities in medical insurance claims. This is important to prevent individuals or organizations from committing fraudulent acts and taking advantage of the healthcare system. The detection process involves analyzing various data sources, such as medical records, billing information, and patient information, to identify any suspicious patterns or behavior.

Zhang et al. (2022) proposed a health insurance fraud detection method that utilizes a hierarchical multi-modal fusion on dynamic heterogeneous graph. The graph includes nodes such as claimer, department, hospital, health care, and timestamp, and edges representing the relationships between them. The goal of the method is to identify fraudulent claimer nodes. On the other hand, Chen et al. (2022) developed a medical insurance fraud detection approach using graph neural networks with spatio-temporal constraints. The graph includes nodes such as patient, fraudster, hospital department, and medicine, and edges representing the relationships between them, such as patient-hospital department-medicine-hospital department-patient. The goal of the method is to detect fraudulent patients. Both papers focus on fraud detection in the medical insurance field.

4.2.2. Datasets

Reviewing papers for the datasets they analyzed and categorizing them from this perspective is advantageous since acquiring an appropriate dataset for research purposes, particularly in the financial domain, has always been a challenging task due to the confidential nature of financial data. In light of this challenge, it was expected to see some private dataset in the studies, especially in certain areas like taxation, where publicly available data was scarce. However, some areas like the area of cryptocurrency have a relatively higher volume of publicly available data. Fig. 10 shows the proportion of each category of dataset used in the reviewed studies (Fig. 11).

There are two types of datasets used in fraud detection research: real-



Fig. 10. Database Proportions in Financial Fraud Detection Applying Graph Neural Networks.

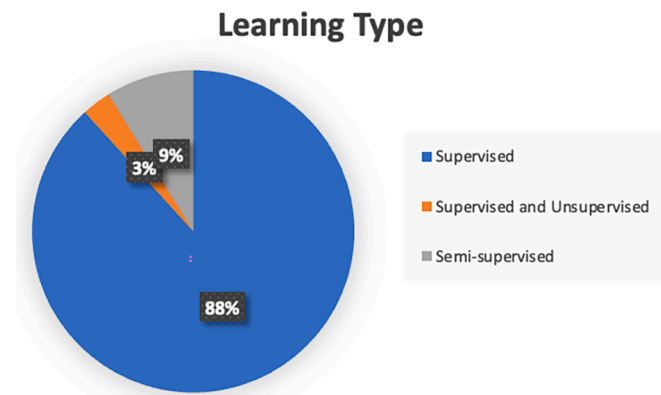


Fig. 11. Learning Mechanism in Financial Fraud Detection Applying Graph Neural Networks.

world and synthetic. Real-world datasets are obtained from actual systems, while synthetic datasets are generated artificially to simulate certain characteristics of real-world data. Among the reviewed papers, only 10 % used synthetic data, while the remaining 90 % used real-world datasets.

Graph datasets can be constructed in two ways: firstly, they can be directly represented as a graph which is a collection of nodes and edges as connections between them, such as (Patel et al., 2022; J. Zhang et al., 2022). Alternatively, the graph datasets can be derived from transactional data by transforming financial transactions into a graph-based format, such as (Van Belle et al., 2022; J. Zhou et al., 2022).

In cryptocurrency area, one of the commonly used datasets in the literature was the Bitcoin Elliptic dataset,⁶ which is accessible through the Kaggle platform. When it comes to detecting fraud in cryptocurrencies, there are also several websites that can be utilized to extract Ethereum data. For instance, xblock.pro/Ethereum, etherscandb.info/scams, and etherscan.io are some of the commonly used websites that provide access to relevant data.

Bitcoin Elliptic dataset contains transactional data of the Bitcoin blockchain network and has been utilized in many studies, seven papers out of 25 papers that made their datasets publicly available, for the purpose of detecting fraud and illicit activities. Transactions in this graph are represented as nodes, while the flow of Bitcoins between two

transactions is represented as an edge connecting them. Each node in the graph is associated with 166 features and has been categorized as having been initiated by a “licit”, “illicit”, or “unknown” entity.

In insurance area, a publicly available dataset related to medical insurance has been made released through a competition⁷ and applied for fraud detection using graph neural networks. In this dataset, the nodes represent the entities involved in the medical insurance claims process such as claimer, department, hospital, health care, and timestamp. The edges represent the relationship between these entities, and the set of edge types is defined as $R = \{(claimer, department), (claimer, hospital), (claimer, health care), (claimer, timestamp)\}$.

Discussing banking data, the sources available consists of a synthetic dataset published on Github,⁸ private real data, and public data on credit card transactions. One example of public data is the German Credit dataset,⁹ which contains information on 1000 credit applicants and their associated attributes, such as job, sex, age, savings accounts, housing, credit amount, checking account, duration, and purpose. In this dataset, the nodes represent clients, and the edges indicate the similarity between two clients.

The use of private or synthetic datasets is common in online transaction studies due to privacy concerns. However, the IEEE-CIS Fraud Detection dataset¹⁰ provides public labeled data on whether online transactions are fraudulent or not. This dataset represents users as nodes and transactions as edges, and includes various features such as payment amount, product code, payment card information, device information, and network connection information of the users involved in the transactions.

Table 7 summarizes the most commonly used datasets, categorizing them into different groups with regard to their financial application fields.

Table 8 provides information on the availability of source codes for the reviewed papers. The table lists the papers and the links to their source codes, if available. Some papers do not have available source codes, while others have links to Github repositories where the code can be accessed. It is important to note that only approximately 15 % of the reviewed papers have provided access to their source codes, which can

⁶ <https://www.kaggle.com/datasets/ellipticco/elliptic-data-set>.

⁷ <https://tianchi.aliyun.com/competition/activeList>.

⁸ https://github.com/Charlesvandamme/Inductive-Graph-Representation-Learning-for-Fraud-Detection/blob/master/Demo/demo_ccf.csv.

⁹ <https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/>.

¹⁰ <https://www.kaggle.com/competitions/ieee-fraud-detection/data>.

Table 7

Datasets analyzed in financial fraud detection using graph neural networks.

Financial Area	Dataset Name	Source	Availability	Real or Synthetic	References
Taxation	TaxSH	A local tax bureau in China	Not Public	Real-world	(Shi et al., 2023)
Cryptocurrency	Bitcoin Elliptic Data	https://www.kaggle.com/ellipticco/elliptic-dat-a-set	Publicly available	Real-world	(Patel et al., 2022; Alarab et al., 2020; Zheng, 2022; Li et al., 2022; Han et al., 2022; Xia et al., 2022; Singh et al., 2021)
	Bitcoin	https://github.com/JetQe/Bitcoin-network-illicit-addresses	Publicly available	Real-world	(Tian et al., 2021)
	Bitcoin Alpha	http://www.btc-alpha.com	Publicly available	Real-world	(Liangguang & Zhen, 2021)
	Ethereum	Public yellow book of Ethereum, https://etherscan.io/	Publicly available	Real-world	(Tan et al., 2021)
	Ethereum	https://www.kaggle.com/xblock/ethereum-phishing-transaction-network	Publicly available	Real-world	(Li et al., 2021)
	Ethereum	xblock.pro/Ethereum	Publicly available	Real-world	(Patel et al., 2020; Patel et al., 2022; Zhou et al., 2022; Duan et al., 2022)
	Ethereum	https://etherscan.io/	Publicly available	Real-world	(Li et al., 2022; Fu et al., 2022; Liu et al., 2022; Li et al., 2022; Li et al., 2022)
Insurance	Medical insurance data	https://tianchi.aliyun.com/competition/entrance/231607/introduction	Publicly available	Real-world	(Zhang et al., 2022)
	Medical insurance data	A private data of a municipal medical insurance bureau in 2018	Not public	Real-world	(Chen et al., 2022)
Banking	German Credit Card Data UC Irvine Repository	https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/	Publicly available	Real-world	(Jing et al., 2019)
	Credit Card Data	https://github.com/Charlesvandamme/Inductive-Graph-Representation-Learning-for-Fraud-Detection/blob/master/Demo/demo_ccf.csv	Publicly available	Synthetic	(Van Belle et al., 2020; Van Belle et al., 2022)
Online Payment Transactions	Credit Card Data	A private data of a bank	Not Public	Real-world	(Cheng et al., 2022)
	CIS-Fraud Data	https://www.kaggle.com/c/ieee-fraud-detection/data	Publicly available	Real-world	(Rao et al., 2021; Sergadeeva et al., 2022)
	Online Purchase History	JD Finance Anti-Fraud dataset	Not public	Real-world	(Lv et al., 2019)
	Consumer Online Loan	Alipay Dataset	Not Public	Real-world	(Xu et al., 2021; Wang et al., 2021)
	Card Transaction Data	Synthesized from a private real-world data	Not Public	Synthetic	(Reddy et al., 2021)
	Alipay	Alipay Website Dataset	Not Public	Real-world	(Wang et al., 2021)

Table 8

Publicly Available Source Codes of the Models Mentioned in the Articles of this Literature Review.

Model	Link	Reference
Apan: Asynchronous propagation attention network	https://github.com/WangXuhongCN/APAN	(Wang et al., 2021)
Ethident: Ethereum account identification framework	https://github.com/jjzhou012/Ethident	(J. Zhou et al., 2022)
EvAnGCN: Evolving Anomaly detection GCN	https://github.com/vatsalpatels/EvAnGCN_v1	(Patel et al., 2022)
OCGNN: One-Class Graph Neural Network-based anomaly detection	https://github.com/vatsalpatels/Dataset_OCGNN	(Patel, Pan, and Rajasegarar, 2020)
Inductive Graph Representation Learning for fraud detection	https://github.com/Charlesvandamme/Inductive-Graph-Representation-Learning-for-Fraud-Detection	(Van Belle et al., 2022)

limit the reproducibility of the studies.

4.2.3. Heterogeneity in graphs

Although many real-world financial systems involve multiple stakeholders and span across various domains, the heterogeneity that arises when attempting to model these larger systems has not been adequately addressed in the current literature. In fact, only around 30 % of studies in the field of graph neural networks have focused on heterogeneous graphs for fraud detection, Bipartite graphs as a type of heterogeneous graph (Reddy et al., 2021; Van Belle et al., 2022), and only a few studies (Shi et al., 2023) have investigated edge-aware heterogeneous graphs. Most of the existing papers have not exclusively used purely heterogeneous graphs, with different types of nodes and edges,

but rather graphs with different types of nodes.

In different heterogeneous graphs, nodes can represent various entities such as production, investors, events, and companies (Shi et al., 2023), users and merchants (Cheng et al., 2022; Reddy et al., 2021; Van Belle et al., 2022; Van Belle, Mitrovic, and De Weerd, 2020), claimers, departments, hospitals, healthcare providers, and timestamps (J. Zhang et al., 2022), or accounts, states, gas and fees, transactions, and blocks (Liu et al., 2022). The edges in these graphs can indicate the type of interaction or relationship between the nodes they connect.

However, as financial systems continue to grow and become increasingly interconnected, there is an emerging need for fraud detection models that can effectively handle graphs with multiple node and edge types. Such models can lead to better generalization in fraud detection, as well as enhanced scalability and cross-domain applicability. By developing graph neural network-based approaches that can effectively manage the complexities of heterogeneous graphs in financial systems, researchers can pave the way for more robust and adaptive fraud detection solutions that can address the evolving challenges of financial fraud in an increasingly interconnected world.

4.2.4. Graph type over time

According to the definitions provided in the preliminaries section, graphs can be classified into three types based on whether they change over time: static, temporal, and dynamic. Static graphs have fixed features and structure over time, while temporal graphs have changing features but the same set of nodes and edges. Dynamic graphs change over time, with the possibility of new nodes and edges being added or removed. Our review of the literature indicates that approximately 50 % of studies focused on static graphs, while around 30 % examined temporal graphs, and the remaining 20 % investigated dynamic graphs.

These findings highlight the need for more research on dynamic

graphs, given that many financial problems involve changes over time. For instance, financial transactions are inherently dynamic processes, as people constantly transfer money between one another, causing the graph structure to change as new users enter the system and engage in transactions. Similarly, online transactions and insurance systems also exhibit dynamic properties. Neglecting to model the temporal and dynamic aspects of these domains may lead to the development of fraud detection algorithms that rely solely on snapshot-based graph analysis.

It is crucial to create a fraud detection model that acknowledges the problem's dynamic nature and accommodates dynamic graphs, particularly in real-world applications where most financial systems operate online and are continuously evolving. By incorporating the dynamic structure of financial systems into graph neural network models, researchers can significantly enhance the accuracy and effectiveness of fraud detection, enabling a more robust response to fraudulent activities as they unfold in real-time.

4.2.5. Learning mechanism

Graph neural networks can be trained for anomaly detection in three different ways: supervised, semi-supervised, and unsupervised. In the supervised setting, labeled data is provided for training. The semi-supervised setting involves a small number of labeled nodes and a large number of unlabeled nodes for training. In contrast, the unsupervised setting only provides unlabeled data for the model to identify patterns.

In our systematic literature review, Shi et al. (2023) presented the sole framework capable of being trained in an unsupervised manner. This paper utilized a *meta*-path based GNN to extract node representations and applied an unsupervised training loss for anomaly detection, and offered a novel evaluation measure to compare models in under unsupervised training.

The unsupervised setting offers a more realistic context for fraud detection, as a significant number of fraud cases may go undetected, necessitating systems with the ability to identify emerging fraud patterns. Solely relying on hand-labelled data could restrict the generalizability of our models and lead to subpar performance when faced with real-world data. This underscores the need for an end-to-end graph neural network approach tailored to unsupervised fraud detection.

4.2.6. Node level vs edge level anomaly detection

Node-level graph anomaly detection involves detecting anomalous behavior or characteristics of individual nodes in a graph. Edge-level graph anomaly detection, on the other hand, involves identifying anomalous behavior or characteristics of the edges in a graph. Based on our study, only %1 of studies experimented models for edge level anomaly detection and rest are focused on node level anomaly detection.

While the definition of nodes and edges can vary depending on the context, in most of the papers reviewed, nodes do not necessarily represent what is on the edges. For example, in (Wang et al., 2021) the definition of nodes, users, does not correspond to the entities on the edges, which are money transactions. However, in a few cases (Van Belle et al., 2020; Xia et al., 2022), each transaction between nodes considered as a node to include embedding of it.

Overall, the literature reviewed in this study reveals that there are few papers that have covered edge level anomaly detection, less than % 10, and no papers that have covered graph level anomaly detection or the detection of abnormal communities in a graph. These types of anomalies are common in the financial industry, such as money laundering or fraudulent groups, and detecting them could have significant implications for financial security.

4.2.7. Percentage of anomaly

Anomaly percentage refers to the proportion of data points in a dataset that are labeled as anomalous or fraudulent. In anomaly detection, the goal is to identify these data points that do not follow the expected behavior or pattern of the majority of the data. The anomaly

percentage can vary based on the application and the level of tolerance for identifying anomalies. Fig. 12 shows that the anomaly percentage varies across different financial areas such as credit card or cryptocurrency having a higher anomaly percentage than medical insurance or taxation or online payment transactions.

Anomalies are generally not common in datasets, as they represent data points that do not conform to the expected behavior or pattern of the majority of the data. The reviewed paper also confirms that with around 70 % of datasets having less than 15 % anomaly and around 45 % of them having less than 2 % anomaly. Some papers have a anomaly percentage of 50 % because they balanced the number of fraud and normal data to solve a balanced classification problem (Chen et al., 2022; Duan et al., 2022; Y. Li et al., 2021). The commonly used Elliptic dataset, <https://www.kaggle.com/datasets/ellipticco/elliptic-data-set>, has a fixed anomaly percentage of 2 %, which explains why many papers use this exact percentage.

Table 9 provides a detailed information of aforementioned sections in comparison of the papers based on their input data, graph type, learning mechanism, and anomaly type.

4.2.8. Number of nodes and edges

This section studies different datasets based on corresponding number of nodes and edges. Fig. 13 provides a comprehensive visualization of the number of nodes and edges in graph datasets used for financial fraud detection. The central histogram displays the distribution of the number of nodes and edges for each dataset, while box plots along the axes offer additional insights into these distributions, including median values and any outliers. The number of nodes is represented on the horizontal axis, and the number of edges on the vertical axis.

Datasets are color-coded based on their respective financial areas, enabling easy comparison and analysis across domains. The intensity of the color corresponds to the frequency of datasets in each bin, offering a clear representation of dataset distribution. The sample size for the distribution is relatively small, and some domains have only one dataset, as depicted in the table. From the figure, there is evident diversity in the number of nodes and edges, as well as the average node degree, which approximates the graph structure in cryptocurrency and online transaction datasets. This diversity allows fraud detection models in this area to be tested on both small graphs and large datasets to evaluate their scalability. However, the same conclusion cannot be drawn for tax datasets, as there is only one dataset in that area.

4.2.9. Research gaps and future direction

One of the significant research gaps identified in our systematic review is the limited focus on unsupervised learning, despite its potential advantages and practicality in real-world fraud detection scenarios. Most often, in practice, labeled data is scarce, and the ability to identify novel and unseen fraud patterns is essential. Unsupervised learning provides a promising solution to these challenges, as it can work effectively without labeled data and detect previously unobserved patterns. Furthermore, we observed a lack of end-to-end unsupervised GNN models in the literature. As a future research direction, we suggest that the development and evaluation of unsupervised and end-to-end unsupervised GNN models should be prioritized, as they have the potential to greatly enhance fraud detection capabilities.

Another area with notable research gaps is the study of dynamic graphs, particularly in online settings, where edges and nodes are added to the graph as a continuous stream of data. Although one work addressed fraud detection for dynamic graphs, it did not focus on online fraud detection, which is a common requirement in many applications, such as online transactions. Moreover, no research has explored the use of cross-domain heterogeneous graphs for financial fraud detection. Addressing these gaps presents an opportunity for future research. Developing GNN models that can efficiently handle dynamic graphs in online settings and investigating the potential of cross-domain heterogeneous graphs can significantly advance the field of financial fraud

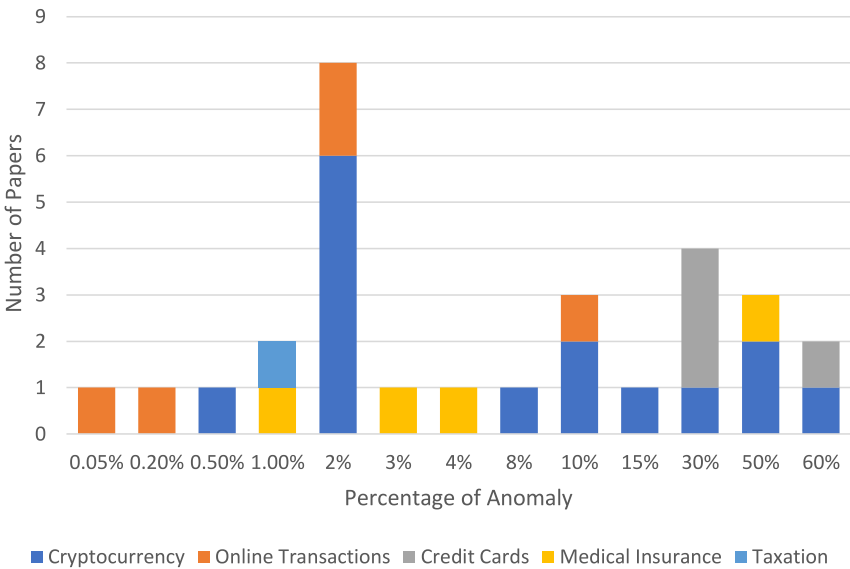


Fig. 12. Number of Papers with Different Anomaly Percentages in Datasets.

Table 9
Comparison of papers according to input, graph type, learning mechanism, and anomaly type.

No.	Screened Papers	Heterogeneity		Graph Type			Learning Mechanism			anomaly Level	
		Homogeneous	Heterogeneous	Static	Temporal	Dynamic	Supervised	Semi-supervised	Unsupervised	Node Level	Edge Level
1	(Patel et al., 2020)	✓		✓			✓			✓	
2	(Patel et al., 2022)	✓				✓	✓			✓	
3	(Wang et al., 2021)	✓				✓	✓				✓
4	(Shi et al., 2023)		✓	✓			✓		✓	✓	
5	(Tan et al., 2021)	✓		✓			✓			✓	
6	(Liangguang & Zhen, 2021)	✓				✓	✓				✓
7	(Alarab et al., 2020)	✓			✓		✓			✓	
8	(Zheng, 2022)	✓			✓		✓			✓	
9	(Reddy et al., 2021)		✓		✓		✓				✓
10	(Li et al., 2022)	✓		✓			✓			✓	
11	(Zhou et al., 2022)		✓	✓			✓			✓	
12	(Li et al., 2021)	✓			✓	✓	✓			✓	
13	(Zhang et al., 2022)		✓			✓	✓			✓	
14	(Tian et al., 2021)	✓			✓	✓	✓			✓	
15	(Han et al., 2022)	✓		✓			✓			✓	
16	(Xia et al., 2022)	✓			✓		✓			✓	
17	(Jing et al., 2021)	✓		✓			✓			✓	
18	(Cheng et al., 2022)		✓		✓		✓			✓	
19	(Li et al., 2022)	✓		✓			✓			✓	
20	(Rao et al., 2021)	✓		✓				✓		✓	
21	(Chen et al., 2022)		✓		✓		✓			✓	
22	(Xu et al., 2021)		✓	✓			✓			✓	
23	(Fu et al., 2022)	✓		✓			✓			✓	
24	(Duan et al., 2022)	✓		✓			✓			✓	
25	(Liu et al., 2022)		✓	✓			✓			✓	
26	(Van Belle et al., 2022)		✓✓		✓		✓			✓	
27	(Van Belle et al., 2020)		✓	✓			✓✓			✓	
28	(Singh et al., 2021)	✓			✓		✓			✓	
29	(Jing et al., 2019)	✓		✓				✓		✓	
30	(Lv et al., 2019)		✓	✓			✓			✓	
31	(Sergadeeva et al., 2022)	✓		✓				✓		✓	
32	(Wang et al., 2021)	✓				✓	✓			✓	
33	(Li et al., 2022)	✓			✓		✓			✓	

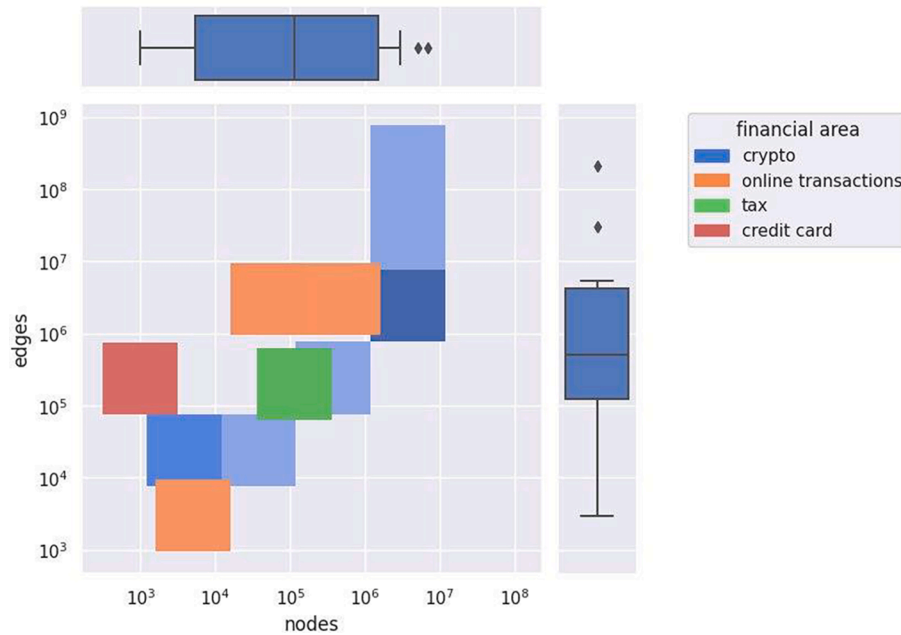


Fig. 13. Number of Nodes and Edges in Different Financial Areas.

detection and broaden its applicability across various financial domains.

5. Threats to validity and limitations

We designed the systematic review and its methodology such that it ensures construct, internal, and external validity as much as possible, while also answering the research questions. However, there are some limitations and validity threats that are mentioned as follows:

- Search query limitations: The search query may not include all the relevant papers; for instance, some papers may use the word “network” instead of “graph” in the title, abstract, or keywords. Although including the word “network” may capture more papers, the query result will increase dramatically with irrelevant papers.
- Non-peer-reviewed articles: This review relies on peer-reviewed literature and neglects the grey literature. Consequently, it might not reflect what is happening in the industry, which is another restriction. Moreover, due to the fast-growing nature of this field, most researchers prefer to publish their papers in early stage of the research through preprint papers, such as ArXiv.
- Generalizability: In terms of generalizability, we expect that the outputs of this study, especially the GNN models, could be employed in any study concerning financial fraud detection. Also, GNN models can also be employed in other applications. However, since the performance of these techniques highly depends on the employed data sources, the techniques may not be suitable for all cases.

6. Conclusion

In this systematic literature review, we thoroughly examined the existing literature and identified relevant papers based on our established criteria for inclusion and exclusion, as well as quality assessment. Our findings contribute to a deeper understanding of the types of graph

neural networks proposed for anomaly detection in financial transactions and the current trends, gaps, and future research directions in this field.

We presented a new taxonomy of GNNs for financial fraud detection that offers a valuable reference for researchers and practitioners. Our analysis of the current trends and gaps revealed the need for further exploration of dynamic and heterogeneous graphs, particularly in cross-domain applications. Moreover, we highlighted the limited availability of diverse graph datasets in some domains, such as taxation, and noted that certain areas of finance, such as real estate and mortgage, to name a few, have not been thoroughly explored.

Although the majority of the papers we reviewed employed supervised methods for fraud detection, we advocate for a more extensive investigation into unsupervised and semi-supervised approaches. These methods are more practical and closer to real-world scenarios, as they hold the potential to reveal previously unseen patterns. This systematic review serves as an up to date reference and a foundation for researchers and practitioners to build upon, as they strive to advance the field of financial fraud detection using graph neural networks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant (RGPIN-2023-03733).

Appendix

Evaluation Metrics

Table A1

Commonly used evaluation metrics, tp: True Positive, tn: True Negative, fp: False Positive, fn: False Negative.

Evaluation Metric	Formula/Description
Accuracy	$\frac{tp + tn}{tp + tn + fp + fn}$
Precision	$\frac{tp}{tp + fp}$
Recall	$\frac{tp}{tp + fn}$
F1 Score	$2 * \frac{Recall * Precision}{Recall + Precision}$
AUC-ROC	The Area Under the ROC Curve
AUC-AP	The Area Under Precision-Recall Curve

Node and Edge Related Features

Table A2

Comparison of papers from graph related properties.

Screened Papers	Fraud Definition	Nodes	Edges	% Anomaly	#nodes	#edges	#Features	Features Examples
(Patel et al., 2020)	Ethereum Attacks related to transactions	The sender and the receiver addresses, i.e., accounts	The set of unique pairs of sender and receiver	36 %	25,257	50,422	34	e.g. block Number, timestamp, transaction Hash, from Address, to Address, value, gas Price, gas Limit, gas Used
(Patel et al., 2022)	Anomalous addresses	Addresses	Connections between addresses	–	–	–	34	e.g. transaction amount, transaction count, transaction average, clustering coefficient
(Wang et al., 2021)	Fraudulent Transactions	Users	Money transferring	–	761,750	2,776,009	101	–
(Shi et al., 2023)	companies with tax evasion	Production, Investor, Event and Company	CC, CP, PC, PP, CE, CI, and II (P: Production, I: Investor, E:Event and C:Company)	0.90 %	112,015	198,903	24	Production code, Education level, Birth place, Age, Address, Event description, Industry, Registration type, Business scope
(Tan et al., 2021)	transaction address engaged in fraudulent transactions on Ethereum.	Addresses	Transaction between a pair of addresses	61 %	3,041	–	2	The total income or expenditure amount and the proportion of the transaction amount to the total transaction amount
(Liangguang & Zhen, 2021)	detect fraudulent transactions	Traders	Transactions	–	3,783	24,186	1	Each edge was labeled with the transaction time.
(Alarab et al., 2020; Zheng, 2022; Li et al., 2022; Tian et al., 2021; Han et al., 2022; Xia et al., 2022; Singh et al., 2021)	Illicit transactions in the Bitcoin transaction graph.	Bitcoin Transactions	Payments flow from source to destination	2 %	203,769	234,355	166	Timestamps, Bitcoin fees, volume, average BTC received/ spent by the inputs/outputs, maximum value, minimum value, standard deviation, and correlation coefficient
(Reddy et al., 2021)	Fraud transactions on a card transaction dataset	Users and merchants	Transactions between users and merchants	0.05 %	62,000	4,000,000	–	Temporal and structural features
(Li et al., 2021)	Phishing accounts	Accounts	Ethereum payment	50 %	1,329,729	2,161,573	–	–
(Zhang et al., 2022)	Fraudulent claimer	Claimer, department, hospital, health care, timestamp	{{(claimer, department), (claimer, hospital), (claimer, health care), (claimer, timestamp)}}	17 % 4.6 % 3.6 %	2,973,489 29,951 16,841	5,355,155	–	nodes have contextual features, e.g. age, gender, and medical expenses;
(Jing et al., 2021)	Fraudulent credit cards	Credit card records	Connections	30 %	1,000	–	24	Each record includes categorial or symbolic characteristics such as job, sex, age

(continued on next page)

Table A2 (continued)

Screened Papers	Fraud Definition	Nodes	Edges	% Anomaly	#nodes	#edges	#Features	Features Examples
(Cheng et al., 2022)	probability of being a fraudulent trade	Users and merchants	Connection between a user and a merchant	–	2,181	235,706	–	–
(Li et al., 2022)	Phishing accounts	Accounts	Transactions with timestamp and amount value	12 %	1,686,003	4,380,616	–	Node feature matrix constructed from the information of transaction frequency
(Rao et al., 2021)	Fraudulent users	Cards	Transactions	–	–	–	–	Cards features such as credit amount, transaction features such as transaction type or amount
(Chen et al., 2022)	Fraudulent patients	Patient, fraudster, hospital department, and medicine.	Relations such as patient-hospital department-medicine-hospital department-patient, ...	50 % 1 %	3,827 8,480	–	–	Average interval between visits, the proportion of multiple hospitals, total personal expenses, fund expenses, and the number of personal visits
(Xu et al., 2021)	Fraudulent Loans	Applicants, sellers, intermediaries and other users	Relation between the applicant and other users	–	–	–	–	Attributes of applicant like user profile, transaction summarizing, and the apply time four different relations of the applicant
(Fu et al., 2022)	Phishing accounts	Addresses (or accounts)	Transactions between addresses	0.50 %	–	–	–	Transaction amounts, transaction fees, number of transactions, and transaction times.
(Duan et al., 2022)	Phishing accounts	Addresses (or accounts)	Transactions between addresses	50 %	3,360	–	–	–
(Liu et al., 2022)	smart contract abnormal detection.	Account, state, gas and fees, transactions and blocks	Relation between nodes	8 %	–	–	–	The account features as characteristics of fraudulent smart contract and the code features as the vectorization of the source code of smart contract.
(Van Belle et al., 2022)	Fraudulent Transactions	Client, merchant, and transaction nodes,	Transactions	0.65 %	–	3,724,348	–	Merchant category code, country, monetary amount, time, acceptance
(Van Belle et al., 2020)	fraudulent transactions	merchants and cardholders, and transactions	connection between merchants and cardholders through the transaction node	0.38 %	4,610,369	–	–	Timestamp information, monetary amount of the transaction, Merchant's country and category
(Jing et al., 2019)	Each record is classified as a good or bad credit risk based on the attributes.	Clients	The edges present the similarity degree of the two clients.	30 %	1,000	499,500	–	job, sex, age, saving accounts, housing, credit amount, checking account, duration and purpose.
(Lv et al., 2019)	fraud user based on users' purchase history information.	Users	users who have shared delivery address, or phone or device, have connection	10 %	5,014	3,000	–	The feature vectors of users contain purchase history of 1000 commodity categories.
(Sergadeeva et al., 2022)	Fraudulent users	Users	Transactions	–	–	–	–	Transaction payment amount, product code, payment card information, users' network connection information
(Wang et al., 2021)	Users who borrows the money from the platform of Alipay cannot return the money on time.	Users	Edge denoting that the user lends the money from the institution.	2 %	5,000,000	30,000,000	553	Amount, the balance, the status, the purpose of the loan, the time to borrow the loan, marriage status, industry.
(Li et al., 2022)	Phishing Addresses	Addresses	Transaction between addresses	~10 %	6,844,050	208,847,461	–	The transaction direction, amount and time information, statistical, structural, trading features.

References

- Abdallah, A., Maarof, M. A., & Zainal, A. (2016). Fraud Detection System: A Survey. *Journal of Network and Computer Applications*, 68, 90–113. <https://doi.org/10.1016/j.jnca.2016.04.007>
- Ahmed, M., Mahmood, A. N., & Rafiqul Islam, M.d. (2016). A Survey of Anomaly Detection Techniques in Financial Domain. *Future Generation Computer Systems*, 55, 278–288. <https://doi.org/10.1016/j.future.2015.01.001>
- Akdoglu, L., Tong, H., & Koutra, D. (2015). Graph Based Anomaly Detection and Description: A Survey. *Data Mining and Knowledge Discovery*, 29(3), 626–688. <https://doi.org/10.1007/s10618-014-0365-y>
- Alarab, I., S. Pragoonwit, and M. I. Nacer. 2020. “Competence of Graph Convolutional Networks for Anti-Money Laundering in Bitcoin Blockchain.” Pp. 23–27 in. Association for Computing Machinery.
- Cai, H., Zheng, V. W., & Chang, K.-C. (2018). A comprehensive survey of graph embedding: problems, techniques, and applications. *IEEE Transactions on Knowledge and Data Engineering*, 30(9), 1616–1637. <https://doi.org/10.1109/TKDE.2018.2807452>
- Chen, J. P., Lu, P., Yang, F., Chen, R., & Lin, K. (2022). Medical insurance fraud detection using graph neural networks with spatio-temporal constraints. *Journal of Network Intelligence*, 7(2), 480–498.
- Cheng, D., Wang, X., Zhang, Y., & Zhang, L. (2022). Graph neural network for fraud detection via spatial-temporal attention. *IEEE Transactions on Knowledge and Data Engineering*, 34(8), 3800–3813. <https://doi.org/10.1109/TKDE.2020.3025588>
- Cherif, A., Badhib, A., Ammar, H., Alshehri, S., Kalkatawi, M., & Imine, A. (2023). Credit card fraud detection in the era of disruptive technologies: A systematic review. *Journal of King Saud University - Computer and Information Sciences*, 35(1), 145–174. <https://doi.org/10.1016/j.jksuci.2022.11.008>
- Duan, X. C., Yan, B. W., Dong, A. M., Zhang, L., & Yu, J. G. (2022). Phishing frauds detection based on graph neural network on Ethereum. *International Conference on Wireless Algorithms, Systems, and Applications* pp. 351–63 in Vol. 13471.
- Duan, X., Yan, B., Dong, A., Zhang, L., & Yu, J. (2022 November). Phishing frauds detection based on graph neural network on Ethereum. In *International Conference on Wireless Algorithms, Systems, and Applications* (pp. 351–363). Cham: Springer Nature Switzerland. Chicago.
- Fu, B., Yu, X., & Feng, T. (2022). CT-GCN: A phishing identification model for blockchain cryptocurrency transactions. *International Journal of Information Security*, 21(6), 1223–1232.
- Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in neural information processing systems*, 30.
- Han, H., Wang, R., Chen, Y., Xie, K., & Zhang, K. (2022). Research on abnormal transaction detection method for blockchain. In *International Conference on Blockchain and Trustworthy Systems* (Vol. 1679, pp. 223–236). CCIS. Springer Science and Business Media Deutschland GmbH.
- Hilal, W., Andrew Gadsden, S., & Yawney, J. (2022). Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances. *Expert Systems with Applications*, 193, Article 116429. <https://doi.org/10.1016/j.eswa.2021.116429>
- Kim, Hwan, Byung Suk Lee, Won-Yong Shin, and Sungsu Lim. 2022. “Graph Anomaly Detection With Graph Neural Networks: Current Status and Challenges.” *IEEE Access* 10:111820–29. 10.1109/ACCESS.2022.3211306.
- Kipf, Thomas N., and Max Welling. 2017. “Semi-Supervised Classification with Graph Convolutional Networks.” In: *Proceedings of ICLR (International Conference on Learning Representations)*.
- Jing, R., Tian, H., Zhou, G., Zhang, X., Zheng, X., & Zeng, D. D. (2021 July). A GNN-based Few-shot learning model on the Credit Card Fraud detection. In *2021 IEEE 1st International Conference on Digital Twins and Parallel Intelligence (DTPi)* (pp. 320–323). IEEE.
- Jing, R., Zheng, X., Tian, H., Zhang, X., Chen, W., Wu, D. D., & Zeng, D. D. (2019 July). A Graph-Based Semi-Supervised Fraud Detection Framework. In *2019 4th IEEE International Conference on Cybernetics (Cyberconf)* (pp. 1–5). IEEE.
- Kitchenham, B. (2004). Procedures for Performing Systematic Reviews. *UK, Keele University*, 33(2004), 1–26.
- Li, A., Wang, Z., Yu, M., & Chen, D. (2022). *Blockchain Abnormal Transaction Detection Method Based on Weighted Sampling Neighborhood Nodes* (pp. 746–752). in. Institute of Electrical and Electronics Engineers Inc..
- Li, Sijia, Gaopeng Gou, Chang Liu, Chengshang Hou, Zhenzhen Li, and Gang Xiong. 2022. “TTAGN: Temporal Transaction Aggregation Graph Network for Ethereum Phishing Scams Detection.” Pp. 661–69 in *Proceedings of the ACM Web Conference 2022*. Virtual Event, Lyon France: ACM.
- Li, Y., Tam, D. S. H., Xie, S., Liu, X., Ying, Q. F., Lau, W. C., ... Chen, S. Z. (2021). Temporal Graph Representation Learning for Detecting Anomalies in E-Payment Systems. In *2021-December, in Vols pp. 983–990*. IEEE Computer Society.
- Li, Yi, Yilun Jin, Guojie Song, Zihao Zhu, Chuan Shi, and Yiming Wang. 2021. “GraphMSE: Efficient Meta-Path Selection in Semantically Aligned Feature Space for Graph Neural Networks.” *Proceedings of the AAAI Conference on Artificial Intelligence* 35(5):4206–14. 10.1609/aaai.v35i5.16544.
- Liangguang, P., & Zhen, L. (2021). *Enhanced temporal graph embedding for identifying fraudulent transactions on transaction networks* (pp. 71–76). Institute of Electrical and Electronics Engineers Inc..
- Liu, L., Tsai, W. T., Bhuiyan, M. Z. A., Peng, H., & Liu, M. S. (2022). Blockchain-Enabled Fraud Discovery through Abnormal Smart Contract Detection on Ethereum. *Future Generation Computer Systems-the International Journal of Esience*, 128, 158–166. <https://doi.org/10.1016/j.future.2021.08.023>
- Lv, L., Cheng, J., Peng, N., Fan, M., Zhao, D., & Zhang, J. (2019 May). Auto-encoder based graph convolutional networks for online financial anti-fraud. In *2019 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFER)* (pp. 1–6). IEEE.
- Ma, X., Jia, W.u., Xue, S., Yang, J., Zhou, C., Sheng, Q. Z., ... Akoglu, L. (2021). A Comprehensive Survey on Graph Anomaly Detection with Deep Learning. *IEEE Transactions on Knowledge and Data Engineering*, 1–1. <https://doi.org/10.1109/TKDE.2021.3118815>
- Niepert, Mathias, Mohamed Ahmed, and Konstantin Kutzkov. 2016. “Learning Convolutional Neural Networks for Graphs.” in *Proceedings of The 33rd International Conference on Machine Learning, PMLR*.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... Moher, D. (2021). The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ*, n71. <https://doi.org/10.1136/bmj.n71>
- Patel, V., Pan, L., & Rajasegarar, S. (2020). Graph Deep Learning Based Anomaly Detection in Ethereum Blockchain Network. In *International conference on network and system security* (Vol. 12570, pp. 132–148). LNCS. Springer Science and Business Media Deutschland GmbH.
- Patel, V., Rajasegarar, S., Pan, L., Liu, J., & Zhu, L. (2022). EvAnGCN: Evolving Graph Deep Neural Network Based Anomaly Detection in Blockchain. In *International Conference on Advanced Data Mining and Applications* (Vol. 13725, pp. 444–456). LNAI. Springer Science and Business Media Deutschland GmbH.
- Rao, Y., Ren, X., Duan, C., Mi, X., Cheng, J., Chen, Y., ... & Wei, X. (2021). Knowledge-guided fraud detection using semi-supervised graph neural network. In *Web Information Systems Engineering-WISE 2021: 22nd International Conference on Web Information Systems Engineering, WISE 2021, Melbourne, VIC, Australia, October 26–29, 2021, Proceedings, Part I 22* (pp. 385–393). Springer International Publishing.
- Reddy, S., Poduval, P., Chauhan, A. V. S., Singh, M., Verma, S., Singh, K., & Bhowmik, T. (2021). *TeGraF: Temporal and Graph Based Fraudulent Transaction Detection Framework*. Association for Computing Machinery Inc.
- Sergadeeva, A. I., Lavrova, D. S., & Zegzhda, D. P. (2022). Bank Fraud Detection with Graph Neural Networks. *Automatic Control and Computer Sciences*, 56(8), 865–873.
- Shi, B., Dong, B., Xu, Y., Wang, J., Wang, Y., & Zheng, Q. (2023). An Edge Feature Aware Heterogeneous Graph Neural Network Model to Support Tax Evasion Detection. *Expert Systems with Applications*, 213. <https://doi.org/10.1016/j.eswa.2022.118903>
- Singh, A., A. Gupta, H. Wadhwa, S. Asthana, and A. Arora. 2021. “Temporal Debiasing Using Adversarial Loss Based GNN Architecture for Crypto Fraud Detection.” Pp. 391–96 in. Institute of Electrical and Electronics Engineers Inc.
- Sun, Y., Han, J., Yan, X., Yu, P. S., & Tianyi, W.u. (2011). PathSim: Meta Path-Based Top-K Similarity Search in Heterogeneous Information Networks. *Proceedings of the VLDB Endowment*, 4(11), 992–1003. <https://doi.org/10.14778/3402707.3402736>
- Tan, R., Tan, Q., Zhang, P., & Li, Z. (2021 December). Graph neural network for ethereum fraud detection. In *2021 IEEE international conference on big knowledge (ICBK)* (pp. 78–85). IEEE.
- Tian, H., Li, Y., Cai, Y., Shi, X., & Zheng, Z. (2021). *Attention-Based Graph Neural Network for Identifying Illicit Bitcoin Addresses* (Vol. 1490 CCIS). Springer Science and Business Media Deutschland GmbH.
- Van Belle, R., S. Mitrovic, and J. De Weerd. 2020. “Representation Learning in Graphs for Credit Card Fraud Detection.” *Mining Data for Financial Applications: 4th ECML PKDD Workshop, MIDAS 2019, Würzburg, Germany, September 16, 2019, Revised Selected Papers*, Pp. 32–46 in Vol. 11985.
- Van Belle, R., Van Damme, C., Tytgat, H., & De Weerd, J. (2022). Inductive Graph Representation Learning for Fraud Detection. *Expert Systems with Applications*, 193. <https://doi.org/10.1016/j.eswa.2021.116463>
- Velicković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph Attention Networks. In: *Proceedings of ICLR (International Conference on Learning Representations)*.
- Wang, X., Lyu, D., Li, M., Xia, Y., Yang, Q., Cui, P., ... Guo, Z. (2021). APAN: Asynchronous Propagation Attention Network for Real-Time Temporal Graph Embedding (pp. 2628–2638). in. Association for Computing Machinery.
- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Philip, S. Y. (2021). A Comprehensive Survey on Graph Neural Networks. *IEEE Transactions on Neural Networks and Learning Systems*, 32(1), 4–24. <https://doi.org/10.1109/TNNLS.2020.2978386>
- Xia, P., Ni, Z., Xiao, H., Zhu, X., & Peng, P. (2022). A Novel Spatiotemporal Prediction Approach Based on Graph Convolution Neural Networks and Long Short-Term Memory for Money Laundering Fraud. *Arabian Journal for Science and Engineering*, 47(2), 1921–1937. <https://doi.org/10.1007/s13369-021-06116-2>
- Xiao, S., Wang, S., Dai, Y., & Guo, W. (2022). Graph Neural Networks in Node Classification: Survey and Evaluation. *Machine Vision and Applications*, 33(1), 4. <https://doi.org/10.1007/s00138-021-01251-0>
- Xu, B., Shen, H., Sun, B., An, R., Cao, Q., & Cheng, X. (2021). May). Towards consumer loan fraud detection: Graph neural networks with role-constrained conditional random field. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(5), 4537–4545.
- Xu, K., Weihua, H.u., Leskovec, J., & Jegelka, S. (2019). How Powerful Are Graph Neural Networks?. In: *Proceedings of ICLR (International Conference on Learning Representations)*.
- Zheng, Y. (2022). GRU-GAT Model for Blockchain Bitcoin Abnormal Transaction Detection. In *IEEE Conference on Telecommunications, Optics and Computer Science (TOCS)* (pp. 666–674).
- Ye, Z.i., Kumar, Y. J., Sing, G. O., Song, F., & Wang, J. (2022). A Comprehensive Survey of Graph Neural Networks for Knowledge Graphs. *IEEE Access*, 10, 75729–75741. <https://doi.org/10.1109/ACCESS.2022.3191784>
- Yildiz, K., Dedebeek, S., Okay, F. Y., & Simsek, M. U. (2022). In *Anomaly Detection in Financial Data Using Deep Learning: A Comparative Analysis* (pp. 1–6). Antalya, Turkey: IEEE.
- Yun, Seongjun, Minbyul Jeong, Raehyun Kim, Jaewoo Kang, and Hyunwoo J. Kim. 2019. “Graph Transformer Networks.” *Advances in neural information processing systems*, 32.

- Zhang, J., F. Yang, K. Lin, and Y. Lai. 2022. "Hierarchical Multi-Modal Fusion on Dynamic Heterogeneous Graph for Health Insurance Fraud Detection." *IEEE International Conference on Multimedia and Expo (ICME)* Vols. 2022-July. IEEE Computer Society.
- Zhang, Z., Cui, P., & Zhu, W. (2022). Deep Learning on Graphs: A Survey. *IEEE Transactions on Knowledge and Data Engineering*, 34(1), 249–270. <https://doi.org/10.1109/TKDE.2020.2981333>
- Zhou, J., Hu, C., Chi, J., Wu, J., Shen, M., & Xuan, Q. (2022). Behavior-Aware Account De-Anonymization on Ethereum Interaction Graph. *IEEE Transactions on Information Forensics and Security*, 17, 3433–3448. <https://doi.org/10.1109/TIFS.2022.3208471>
- Zhou, J., Cui, G., Shengding, H.u., Zhang, Z., Yang, C., Liu, Z., ... Sun, M. (2020). Graph Neural Networks: A Review of Methods and Applications. *AI Open*, 1, 57–81. <https://doi.org/10.1016/j.aiopen.2021.01.001>
- Zhou, Y.u., Zheng, H., Huang, X., Hao, S., Li, D., & Zhao, J. (2022). Graph Neural Networks: Taxonomy, Advances, and Trends. *ACM Transactions on Intelligent Systems and Technology*, 13(1), 1–54. <https://doi.org/10.1145/3495161>
- Zhu, X., Ao, X., Qin, Z., Chang, Y., Liu, Y., He, Q., & Li, J. (2021). Intelligent Financial Fraud Detection Practices in Post-Pandemic Era. *The Innovation*, 2(4), Article 100176. <https://doi.org/10.1016/j.xinn.2021.100176>